



Federal Office
for Information Security

BSI Technical Guideline TR-03135-1

Machine Authentication of MRTDs for Public Sector Applications

Part 1: Overview and Functional Requirements

Version 5.0.1



Federal Office for Information Security
PO Box 20 03 63
53133 Bonn
E-Mail: tr-03135@bsi.bund.de
Internet: <https://www.bsi.bund.de>
© Federal Office for Information Security

Table of Contents

1	Changelog Part 1	1
1.1	Version 5.0.1	1
1.2	Version 5.0.0	1
1.3	Version 2.5	2
2	Introduction	3
2.1	Structure of this Technical Guideline	3
2.2	Technical terms	3
2.3	Terminology	3
2.4	Publication Language	3
3	Overview	4
3.1	Document components	4
3.2	MRTD features	5
3.3	Document reader component	5
3.4	Inspection Application	5
3.5	Inspection System	5
3.6	Authentication processes and their result	5
3.7	Different types of checks	7
3.8	Operational profiling	8
4	Inspection System Requirements	10
4.1	Reader requirements	10
4.2	Background Public Key Infrastructures Requirements	11
4.3	Requirements for comparing 2-letter with 3-letter country codes	12
5	Authentication of MRTDs	14
5.1	Process sequence of checking an MRTD	14
5.2	Visualization of check results	16
5.3	Error codes	16
5.4	Overall document check result	16
5.5	Aggregation of overall check results	17
5.6	Common checks	17
5.7	Checking an optical component	30
5.8	Checking an electronic component	51
5.9	Checking a digital seal component	75
5.10	Combined checks	80
5.11	Cross document checks	85
5.12	Handling and interpretation of defects	87
6	Logging scheme for operational monitoring	91
6.1	Use cases	91
6.2	XML schemas	91
6.3	Logging profiling	92
6.4	Sequence of events	92

7	Conformity	98
	List of Abbreviations	99
	Bibliography	101

List of Figures

4.1. Example on how to place a passport and an ID card on a typical document reader	10
4.2. Schematic representation of the architecture IS <-> DV <-> N-PKD <-> CSCA resp. DV <-> SPOC <-> CVCA	12
5.1. Process sequence of checking MRTDs	15
5.2. Example sequence of optical checks of MRTDs	34
5.3. Overview of electronic chip access process	68
5.4. Subprocess "Chip Access"	69
5.5. Subprocess "Perform PACE"	70
5.6. Subprocess "EACv2"	71
5.7. Subprocess "EACv1"	72
5.8. Subprocess "Process data groups without TA"	72
5.9. Subprocess "Process data groups with TA"	72
6.1. Temporal relation of optical, electronic, digital seal and combined checks	97

1 Changelog Part 1

The following tables present the changes introduced between the latest versions of this Technical Guideline. The changelog lists the changes grouped per part of this Technical Guideline, per element (section, table, figure) and type of change (refer to [KeepAChangelog]):

- *Added* for new features
- *Changed* for changes in existing functionality
- *Deprecated* for soon-to-be removed features
- *Removed* for now removed features
- *Fixed* for any bug fixes
- *Security* in case of vulnerabilities

1.1 Version 5.0.1

Element Name	Type of Change	Change Description
►Section 5.8.3	Changed	Chip access protocols results are now included in the electronic check and the aggregation table of access protocols check results is updated.
►Section 5.8.2	Added	Check for the presence of applications
►Section 5.7.3.7	Added	Check for the presence of the primary facial image
►Section 5.7.3.5	Added	Check if the document is only valid in the issuing country.
►Section 5.6.1.2.11	Added	Rule for estimating the year of birth from the MRZ added.
►Section 5.10.2	Changed	Rule for estimating the document's expiration date changed.
►Section 5.12.2.1.3	Changed	Reference to section ►Section 5.10.2 added.

Table 1.1 Changelog Version 5.0.1

1.2 Version 5.0.0

Element Name	Type of Change	Change Description
►Table 3.1	Added	Explicit rules for aggregation of multiple check results were added.
►Table 5.1	Changed	Recommended traffic light colour for the check result "aborted" was changed from grey to black.
►Table 5.1	Added	Recommended pictograms were added for the visualization of the check results.
►Table 5.3	Added	Aggregation table for overall check result was added.
►Table 5.39	Added	Aggregation table for optical checks was added.
►Table 5.80	Added	Aggregation table for electronic checks was added.
►Table 5.131	Added	Aggregation table for combined checks was added.
►Table 5.119	Added	Aggregation table for digital seal checks was added.
►Section 5.8.10	Changed	The figures with processing sequences were replaced with Business Process Model and Notation (BPMN) diagrams. The corresponding text was rewritten.
►Section 4.3	Added	This section was added to describe the comparison of 2-letter and 3-letter country codes only once for usage in different contexts (optical component, electronic component, digital seal).
►Section 5.6.1	Added	This section was added to define MRZ data checks only once for usage in different contexts (optical component, electronic component, digital seal).
►Section 5.6.3	Added	This section was added to define certificate checks only once for usage in different contexts (electronic component, digital seal).

Element Name	Type of Change	Change Description
►Section 5.6.2	Added	This section was added to define issuing country checks regarding signer certificates only once for usage in different contexts (electronic component, digital seal).
►Section 5.7.3	Changed	This section was extended with new MRZ checks (semantical and syntactical) and alignment checks.
►Section 5.7	Added	This section was extended with new checks regarding VIZ.
►Section 5.8.12	Added	Introduction of Forensic Analysis Information List (FAIL) was added.
►Section 5.8.13	Added	Introduction of Digital Travel Credentials (DTC) was added.
►Section 6.2	Changed	Bumped schema files' version
►Section 6.2	Removed	Reference to bc (background check) schema file
►Section 6.2	Removed	Reference to biometric checks schema file
►Section 6.2	Changed	Updated section to reflect current schema files
►Section 6.4	Added	Added section to describe the usage of timestamps.
►Section 5.7	Changed	- Removed vendor specific checks. - Added dedicated checks as replacement. Note: New dedicated checks will be added if reasonable requirements become known.
►Section 5.7	Removed	Removed section Catalogue for spectrally selective checks
►Chapter 2	Added	Added section on publication language
►Section 5.12.2.1.2	Added	Added Signer Certificate Revoked ([BSI TR-03129-2]) for checks using BSI TR-03129-2 version 1.4.2. Replaces Document Signer Certificate Revoked.
►Section 3.8	Added	Added Conditionally Ignored

Table 1.2 Changelog Version 5.0.0

1.3 Version 2.5

Element Name	Type of Change	Change Description
Chapter Authentication of MRTDs	Added	New section "Digital Seal Checks" to support the checking of digital seals
Chapter Authentication of MRTDs	Added	New combined checks related to digital seals
Logging scheme for operational monitoring	Changed	Updated schema versions
Logging scheme for operational monitoring	Added	New dcs schema

Table 1.3 Changelog Version 2.5

2 Introduction

Machine Readable Travel Documents (MRTDs) usually feature a standardized Machine Readable Zone (MRZ) with biographic data and an electronic chip. In addition to typical textual areas on MRTDs, many documents also feature special optical/physical properties which can only be recognized and evaluated by special recording equipment. These properties are intended as an additional indicator for the authenticity of a document or they can be used for establishing a link between the document and its owner. Besides these features, the integrated chip contains information that is printed on the document, e. g. name, date of birth, but also biometric characteristics and cryptographic material that ensures that a given document was issued by an official entity and was not altered, cloned or forged.

This Technical Guideline (TR) defines requirements for document inspection using document inspection systems and a normative description of the necessary procedures that are required for an exhaustive check of (electronic) machine readable identity and travel documents, to acknowledge that a checked MRTD is authentic, of integrity, and valid.

The German Federal Office for Information Security (Bundesamt für Sicherheit in der Informationstechnik (BSI)) composed [BSI TR-03135] (TR : Technische Richtlinie, German for Technical Guideline) in collaboration with the Federal Criminal Police Office (Bundeskriminalamt (BKA)), the Federal Police (Bundespolizei (BPOL)), and the Federal Office of Administration (Bundesverwaltungsamt (BVA)).

2.1 Structure of this Technical Guideline

This Technical Guideline consists of the following parts:

- Part 1: Overview and Functional Requirements (TR-03135-1)

This part describes the process and protocols for reading Electronic Machine Readable Travel Documents (eMRTDs). It outlines the optical, electronic, and combined checks used to ascertain the authenticity, integrity, and validity of such documents. It also provides instructions and procedures for handling of defects which may occur during an inspection process. Furthermore, it specifies a logging scheme for recording the technical activities and results of an inspection process.

- Part 2: Application profiles for official document inspection systems (TR-03135-2)

This part describes the operational scenarios for document inspection systems in public sector applications, as well as the corresponding requirements regarding document checks and operational monitoring.

- Part 3: High Level Document Check Interface Specification (TR-03135-3)

This part describes a client-server architecture for document checks. It specifies interfaces for document checks controlled by workflow definitions and transaction-based logging of the results.

2.2 Technical terms

To improve the reading experience terms are abbreviated. The abbreviations are explained in the ►glossary.

2.3 Terminology

Keywords used within in this document are to be interpreted as described in [RFC2119].

2.4 Publication Language

This Technical Guideline is only published in English. The reasons for this are as follows:

- Relation to other English only publications (esp. [ICAO9303], [BSI TR-03129-2], [BSI TR-03121] and [BSI TR-03110])
- The content is relevant in a European context.

3 Overview

This chapter describes the basic architecture of document inspection environments. Requirements for the document readers are described in ▶Chapter 4, while the coverage of such checks are specified in ▶Chapter 5. ▶Chapter 6 specifies the contents and the logging format of the document checking process. Conformance requirements are described in ▶Chapter 7.

It is recommended to have a look into “Machine Readable Travel Documents” [ICAO9303] and “Best Practice Operational Guidelines for Automated Border Control (ABC) Systems” [FRONTEX]. For Identity Documents the Technical Guidelines [BSI TR-03127] and [BSI TR-03110] are recommended as additional reading.

3.1 Document components

This TR views an MRTD as a set of several components. The types of components are as follows:

- Optical component (see ▶Section 3.1.1),
- Electronic component (see ▶Section 3.1.2),
- Digital seal component (see ▶Section 3.1.3).

Each component consists of several features, which can be processed by machines. The features are recorded by a document reader component (see ▶Section 3.3) and processed in an Inspection Application (see ▶Section 3.4). Internationally formalized and standardized document features of MRTDs are specified in [ICAO9303]. In addition to standardized features, many documents also contain specific information and features which can be used as an indicator for the authenticity and integrity of a specific MRTD. MRTD features can be catalogued, so software can rely on such information to perform a complete and reliable document authentication.

However, not all features of a MRTD have to be checked or can be checked. The scope of the check is dependent on the application scenario (see [BSI TR-03135-2]), the MRTD type and the Inspection System (IS) (see ▶Section 3.5) used to read and analyse the features. It is RECOMMENDED to specify mandatory checks in a profiling document with respect to this Technical Guideline.

3.1.1 Optical component

In general, the document under inspection is a physical document with pages that can be checked optically. These contain printed data regarding the holder and issuer of the document. The most prominent way to transport information for computer assisted processing is by encoding these inside a machine-readable string, the MRZ, which can be read using Optical Character Recognition (OCR). Besides, information that is human readable is printed on the document as well. In principle, this information can be read automatically, however, this process is more prone to errors since no special font is used and there are no check numbers. Printed documents are protected by using special optical patterns, paper and dyes and printing techniques to prevent tampering. In the context of this TR, an optical component encompasses all the document pages required for an inspection.

3.1.2 Electronic components

More advanced documents, such as passports, can contain a chip that stores information which is cryptographically protected against modification and against physical tampering. Additionally, access to the information as well as the reading process can be protected. Besides on chips, information can also be stored on other authenticators, such as smart devices, in case of Digital Travel Credentials (DTCs). In both cases, chip and DTCs, are regarded as electronic components of a document.

3.1.3 Digital seal components

Barcodes or multi-dimensional matrix codes are methods to encode information in a machine readable format. They can also be used to protect the information contained against tampering. Though, in general, less information can be stored when compared to electronic components (e.g. chips). A document can encompass several digital seal components. Each of them can be checked as part of a document check.

3.1.4 Checking components

Inspecting a document means checking its components. Besides individual checks, this TR defines checks between the components. These combined checks are used to compare the information and check for plausibility.

Depending on the checking process, more than one document needs to be processed in an inspection. In these cases, the documents are linked and need be cross checked for plausibility. Here, this is the case for visas.

In the following sections, the document checks of an individual document as well as checks across documents are the subject of discussion.

3.2 MRTD features

This Technical Guideline assumes that MRTDs are equipped with a combination of

- optical/physical and
- electronic (digital)

features, which can be processed by machines. These are recorded by some kind of document reader and processed in an Inspection Application. Internationally formalized and standardized document features are specified in [ICAO9303] for MRTDs. In addition to standardized features, many documents also contain specific information and features which can also be used as an indicator for the authenticity and integrity of a specific MRTD. They can be catalogued, so software can rely on such information to perform a complete and reliable document authentication.

Not all features of a document have to be checked or can be checked. The scope of the check is dependent on the application scenario, the document type and the equipment used to read and analyse the features. It is RECOMMENDED to specify mandatory checks in a profiling document with respect to this Technical Guideline.

3.3 Document reader component

The document reader component, also called document reader, records the features of a given MRTD according to its type. It transmits the collected records to an Inspection Application for further processing.

3.4 Inspection Application

In this Technical Guideline, the Inspection Application controls the reading process of the document reader (component) and processes the information obtained from a MRTD depending on the profile and the type of document. The Inspection Application is able to configure the document reader (component) according to the operational requirements and type of the document to read. It actively monitors and controls the steps needed to exhaustively read and process a given travel document and its features.

3.5 Inspection System

The Inspection System comprises of the Inspection Application, the document reader component and all background information systems involved in the execution of a document check.

3.6 Authentication processes and their result

In the context of this Technical Guideline the main focus is to determine if a given travel document is

- authentic (i. e. genuine, originating from the issuer)
- of integrity (without any manipulation) and
- valid (e. g. not expired, not reported as lost or stolen, not a fantasy document)

These checks can be applied to the entire document or just partial areas of the MRTD and refer to the document's components and their features. An individual check consists of a well-defined checking process. Examples for checking processes are:

- Check if a given document's optical component can be read in near-infrared light (infrared (IR)) wavelength range ~900 nm.
- Check if there are defined patterns of an optical component that become visible or invisible under exposure by a certain light source.
- Check the validity of a document (e. g. via an optical component's MRZ's date of expiry).
- Perform a query to background systems regarding the data of an optical component's MRZ to determine if the document is marked as lost or stolen.
- Calculate the checksums of the data groups read from an electronic component to confirm its integrity.
- Check the digitally signed checksum an electronic component or digital seal component to prove integrity.
- Do a plausibility cross-check between information for an optical component (e. g. MRZ on a page) and the corresponding information from an electronic component (e. g. EF .DG1 in the chip).

An individual checking process SHALL always lead to a well-defined and verifiable result. As such, an individual checking process is either

processed completely according to its algorithm until it is terminated and delivers a well-defined check result or

aborted ahead of schedule and does not deliver a meaningful check result.

The amount of checking processes performed depends on the application scenario defined by the profiling and the work-units specified therein which are prescribed as a complete check.

At the end of the checking processes, all individual checks are assorted to intermediate results and an overall check result for each category to build a final assessment.

3.6.1 The MRTD Checking Process

Within this document, the organized and oriented sequence of a check is formally defined as a checking process. This sequence defines what is to be recorded, measured and controlled as part of its check and which results a process can return.

Remark: In this Technical Guideline, the checking process is just a formally variable and does not define how a check is technically implemented in hard- or software. The description of the technical design and implementation of a checking process, its in- and output values are out of scope.

A checking process P formally describes the sequence of causal dependence state transitions at times $t_1, \dots, t_i, t_{i+1}, \dots, t_E$, i. e. the sequence of the checking events from the start t_1 to the end t_E . Here, a checking process is a series of deterministic state transitions $Z_i \rightarrow Z_{i+1}$ from state Z_i to state Z_{i+1} , whereas each state depends on previous states and is explicitly derived by them.

A checking process can only be in one of the following global states:

1. Initiated

The checking process has not started yet, there has been no activity. The process is in the state Z_0 , the so called idle state.

2. Running

At time i , the checking process is in the state Z_i and is being performed at this time.

3. Terminated

The checking process was performed until the deterministic end (time E) of its life cycle with final state Z_E having been reached. It delivers a sound result that can be used to rate a given document and its checked feature(s).

4. Aborted

The checking process was aborted prior to its defined end at time X of the life cycle with state Z_X . The checking process was not able to finish correctly and thus does not deliver a complete result¹. It is RECOMMENDED that an error code and a short description is provided.

The result of a process terminated at time E and state Z_E SHALL be used for determining the authenticity, integrity and validity. Errors during the operation, hardware, software or handling errors can cause a premature abortion of the process in a state Z_X at time X (i.e. aborted). In order to ensure constant checking quality, only check results arising from a terminated process (terminated at time E and state Z_E) SHALL be used as a check result. The information of any other state (notably state aborted) MAY only be documented and used in some kind of operational monitoring; these values, however, SHALL NOT form a basis for any reasonable check result.

Only the result of a terminated process (terminated at time E and state Z_E) SHALL be input for another basic checking process or any final result.

3.6.2 Check results

The result of a checking process has exactly one of the following results:

- successful

The check **terminated**, the feature of the document to be checked successfully passed the check, i.e. a specific check was able to be applied to the document and assessed according to its specifications

- failed

The check **terminated**, the feature of the document to be checked did not successfully pass the check; in other words, the applied check failed on the feature. The document SHOULD be subject to a further (manual) check for authenticity, integrity and validity

- undetermined

The check **terminated**, it was not possible to determine if the feature to be checked is authentic, of integrity or is valid. The document SHOULD be subject to a further (manual) check.

- not supported

The check **terminated**, but the specific document does not support the features to be checked (for example, a document without a chip, a document which does not have specific security features per definition, a specific protocol was not supported by a chip etc.)

The final result SHALL be given according to one of the values prescribed above. Besides the check result, further information MAY also be attached. The result can then be used together with its attached information in further checks.

Otherwise the check must be **aborted** and there is no meaningful result with regard to the intended document check. In those cases, the following check result shall be used:

- aborted

The check aborted prior to its defined process flow. Obtained check information and results SHOULD NOT be used as proper document check indications.

3.7 Different types of checks

Checks on MRTDs can be divided into several categories, which are:

- Optical checks: These checks concern an optical component.
- Electronic checks: These checks concern an electronic component.

¹ Aborted occurs due to an error while executing the check process and has nothing to do with a check result. For example, if the MRTD was removed too early, before it was completely read and checked, or if a light source is damaged, the scanner sensor does not work properly, the Radio-Frequency Identification (RFID) near field could not be established.

- **Digital seal checks:** These checks concern a digital seal component.
- **Combined checks:** These checks concern multiple components (e.g. comparing an optical component's MRZ data with an electronic component's MRZ data).
- **Cross document checks:** These checks concern separate related documents (e.g. a passport and a visa for the same person).

Typically, further checks are subsequently performed, for example:

- **Biometric checks:** Checking the identity of the document owner based on biometric characteristics (e. g. facial image, fingerprint images) initially recorded when the official document was applied for. These are checked against the live data collected on site to verify that the current document holder is the one the document was issued for.
- **Requests into background systems:** Automated requests to background information systems on the basis of the information featured on an MRTD (e. g. searching with MRZ data in official databases like INPOL, SIS, VIS, RTP, etc.).

Although the document owner's identity will be determined in a complete checking process, this Technical Guideline focuses on checking the document itself. The relation between the owner of an MRTD and the document itself is often established by comparing biometric characteristics (the facial image or fingerprints) to the characteristics stored in the electronic chip of the eMRTD. Checking the biometric characteristics can also be performed manually or by using automated systems. More details and requirements for such checks and logging formats are defined in [BSI TR-03121] and are not part of this TR.

3.8 Operational profiling

This Technical Guideline contains a number of checks that could be performed on MRTDs. The requirements on how and what to check depends on the application scenario and **SHOULD** be specified by the agency responsible for checking MRTDs. As such, all necessary checks and requirements **SHOULD** be defined in a profiling document² referencing this Technical Guideline. Any restrictions, deviations and specific designs of the checks which were not implemented in accordance herein **SHALL** be concretized accordingly in the profiling.

Document checks as defined in this Technical Guideline can be:

- **Mandatory (m)** i.e. **SHALL** always be performed and **SHALL** be included in the final result.
- **Conditionally Ignored (c)** i.e. **SHALL** always be performed and **SHALL** be included in the final result, unless certain conditions are met.
- **Optional (o)** i.e. **MAY** be implemented and performed. If they are implemented in a checking process, the results of the checks **SHALL** be included in the final result.
- **Evaluatory (e)** i.e. **SHALL NOT** be included in the final result and are only documented in the log for subsequent evaluations and operational monitoring.

Depending on the requirement and the profiling, all data obtained during an authentication process **SHALL** be stored according to the monitoring log-data format. Further information and details on this can be found in ►Chapter 6.

Document checks as defined in this Technical Guideline can be aggregated with each other in order to convey the check results at different levels of abstraction and technical depth. ►Table 3.1 provides a generic ruleset for the aggregation of check results, provided that they are to be included in the final result according to their respective run modes as described above. Each entry of this table shows the result of combining the result of check A, which can be found in the first column of the respective row, with the result of check B, which can be found in the first row of the corresponding column.

The aggregation is carried out according to the following order of precedence based on the relative importance of the check results w.r.t. the authentication of eMRTDs:

```
failed > aborted > undetermined > successful > not supported
```

² For example, the German implementation is given as Part 2 of this Technical Guideline.

In other words, aggregating multiple checks yields the most "severe" check result according to the aforementioned order of precedence.

This generic ruleset applies to most aggregations of check results in this TR; however, exceptions where more complex aggregation rules are necessary do exist.

Result of check A (column) - Result of check B (row)	failed	aborted	undetermined	successful	not supported
failed	failed	failed	failed	failed	failed
aborted	failed	aborted	aborted	aborted	aborted
undetermined	failed	aborted	undetermined	undetermined	undetermined
successful	failed	aborted	undetermined	successful	successful
not supported	failed	aborted	undetermined	successful	not supported

Table 3.1 Rules for the aggregation of two check results

4 Inspection System Requirements

4.1 Reader requirements

4.1.1 Document formats

A full-page MRTD document reader SHALL support the sizes and formats for MRTDs specified in [ICAO9303]. This includes documents in the sizes TD-1, TD-2 and TD-3 as well as machine-readable visas in the MRV-A and MRV-B formats ¹.

4.1.2 Reading the MRZ/CAN

The document reader SHALL support automated reading of the MRZ according to the definition in [ICAO9303]. If it is a full-page reader, it SHALL be able to record the image in the near-infrared light (B900 band, 900 +/- 50 nm) and SHOULD be able to read images in the visible wavelength range. More detailed information can be found in [ISO1831].

A full-page reader SHALL also support the automated reading of the six-digit Card Access Number (CAN) that can be found on some eMRTDs or IDs.

It is RECOMMENDED that swipe-type document readers also support reading the CAN in addition to the MRZ.

4.1.3 Applying and handling the document on a full-page reader

MRTDs SHALL be placed on a full-page reader such, that the personalized data page can be recorded optically without any interference.

The MRZ or the CAN are placed in the direction of the document reader. ►Figure 4.1 shows an example on how an electronic passport and an identity card SHOULD be placed on a full-page reader. In the following this will be called the end position of the document.

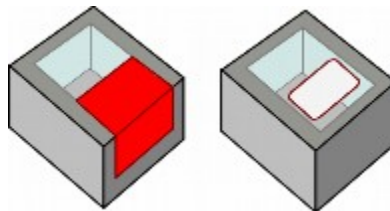


Figure 4.1. Example on how to place a passport and an ID card on a typical document reader

The document reader can digitize the data page of an MRTD and then start further checking steps. It is RECOMMENDED that the electronic reading process should start as soon as the MRZ or the CAN is available. It is RECOMMENDED that digitized records of the extracted MRZ or CAN are available to the system in less than 5 seconds after the document is in the end position (see ►Section 4.1.3).

4.1.4 Communication requirements for the RF-reading module

The Radio-Frequency (RF)-reading module SHALL communicate with a semi-conductor device embedded in the eMRTD using radio frequency energy that meets the requirements specified in [ISO14443-2].

The document reader SHALL be able to read eMRTDs with an electrically conductive shielding.

Reading the electronic data ² of an eMRTD SHALL NOT take longer than 7 seconds on a reference eMRTD, using the minimal system requirements as specified by the vendor of the inspection system. The start time for measuring this requirement is the point in time when the document is in the end position (see ►Section 4.1.3) on the reader and all information needed to process the document (MRZ, CAN) is available to the inspection system.

¹ The document reader SHALL be able to process documents surrounded by a cover that exceeds the data page on all sides up to 5 mm. For example, this applies to old German passports or passports in a slipcover.

² At least the data groups EF.COM, EF.SOD, DG1 and DG2.

4.1.5 Host system interface

If the Inspection System is partitioned in several components, such as document reader, RF-reader, and Inspection Application on a computing unit, the connection interfaces between the components SHOULD preferably be used by means of common industry standards, e. g. Universal Serial Bus (USB).

4.1.6 Requirements for the optical reading component

Images from the MRTDs data page SHALL be optically recorded and digitized under near-infrared light (in the B900 band, cf. document specification according to [ISO1831]), under ultraviolet (UV) light of a wavelength of approx. 365 nm and under white light reflecting colours of the visible spectrum in roughly equal amounts.

Ambient light can have a significantly adverse effect on the quality of the scanned records; thus, it is RECOMMENDED to use a document reader which is insensitive to incoming ambient light or can be shielded.

The document Inspection Application SHALL provide all records of the data page in computer readable image file formats (Portable Network Graphics (PNG) or JPEG (JPEG)) with an image resolution of at least 385 pixels per inch (PPI)³, in the sense of scale mapping the sensor size to the document size. For JPEG, the operator SHALL be able to configure the encoding for different compression factors and quality settings, see [ISO10918-1].

4.2 Background Public Key Infrastructures Requirements

While inspecting the electronic components and digital security features of eMRTDs standardized protocols SHALL be performed. Most of these protocols depend on Public Key Infrastructures (PKIs). These include the protocol that verifies that the data stored on the MRTDs was not altered and is trustworthy and the protocols that detect cloned chip contents.

The PKIs used within this document are issuer PKIs according to [ICAO9303] and Extended Access Control (EAC) PKIs according to [BSI TR-03110] who are fundamental building blocks for a chain of trust regarding eMRTDs.

In order to facilitate a complete electronic check, an Inspection System SHOULD support:

- Implementation of Terminal Authentication (TA) (see ▶Section 5.8.11)
- Master List and Signer List handling and verification of signer certificates and Country Signing Certificate Authority (CSCA) certificates (see ▶Section 5.6.3.2)
- Defect Lists handling for a given Document Signer (DS) or CSCA certificate (see ▶Section 5.12)
- Implementation of Passive Authentication by means of the IS (see ▶Section 5.8.6 and ▶Section 5.8.7)

The profiling SHALL specify what to implement and specify additional details where needed.

4.2.1 Receiving Master Lists, Defect Lists, Document Signer Lists, Signer Lists and access authorization certificates

The components of ISs exchange several lists. A schematic representation of this architecture is given in ▶Figure 4.2.

³ A resolution of 600 PPI is RECOMMENDED.

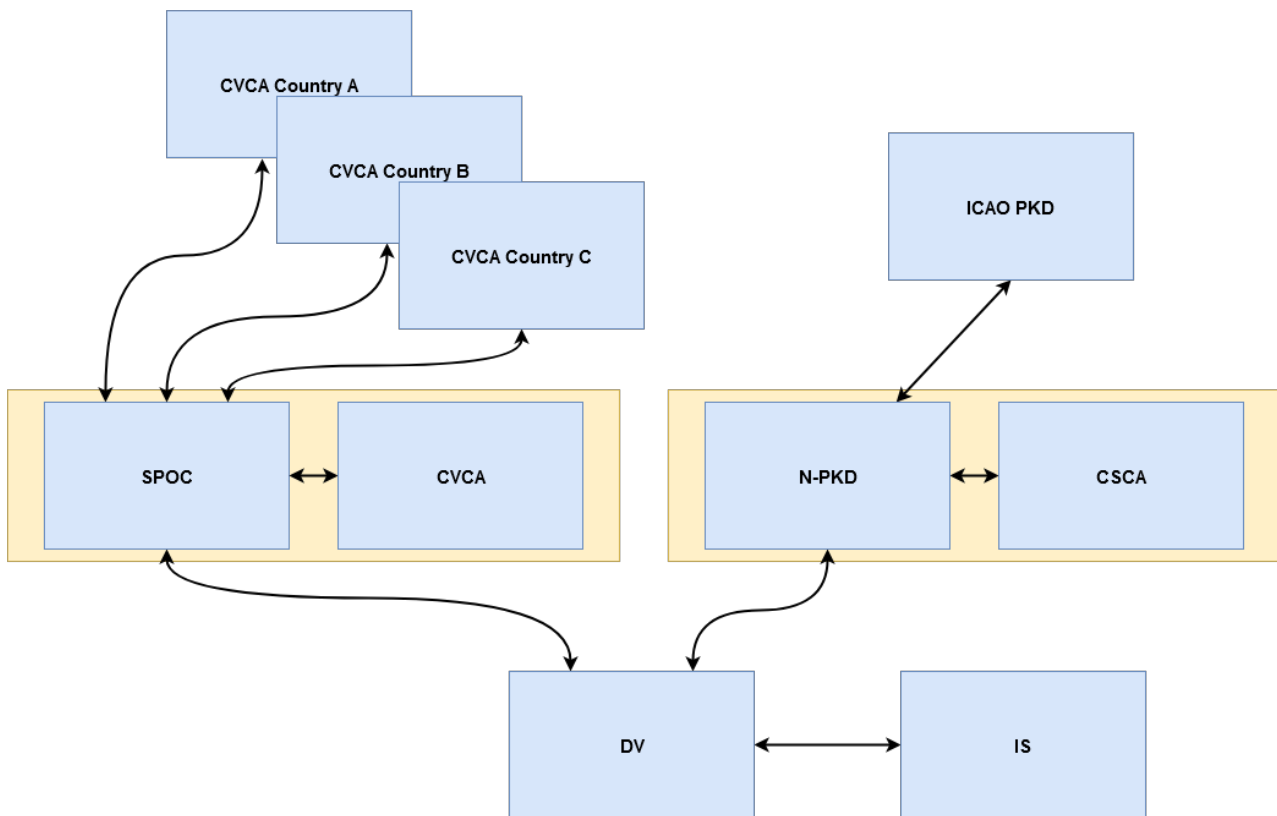


Figure 4.2. Schematic representation of the architecture IS <-> DV <-> N-PKD <-> CSCA resp. DV <-> SPOC <-> CVCA

The IS receives the Master Lists, Defect Lists, Document Signer Lists, Signer Lists (also see [BSI TR-03129-2]), and access authorization certificates via the Document Verifier (DV) which, in turn, is connected to the national Country Verifying Certificate Authority (CVCA) via the Single Point of Contact (SPOC). The national CVCA is connected to other countries' CVCA's via the SPOC. The national Public Key Directory (PKD) is connected to the International Civil Aviation Organization (ICAO) PKD and the national CSCA – it manages the national and international CSCA, DS and Signer certificates which are distributed through Master, Document Signer and Signer Lists. Defects are distributed using Defect Lists.

An online connection between IS and DV is not mandatory in all application scenarios; it is, whenever the necessary certificates or defects SHALL be processed contemporary. With regard to EAC-PKI, an online connection is RECOMMENDED.

Possible implementations for the infrastructure of Inspection Systems are specified in [BSI TR-03129-2], whereby the architecture on the basis of a distributed Inspection System according to [BSI TR-03129-2] SHOULD be used in Inspection Systems for Public Sector Applications, such as border controls and for document checks of the police.

4.3 Requirements for comparing 2-letter with 3-letter country codes

There are various components in the document that identify the issuing country, e.g. in the MRZ or digital certificates in chips, seals, etc. However, due to different specifications for the components, the document issuing country might be represented by country codes of different lengths. Nevertheless, certain checks defined in this TR require to compare the issuing country of the document components. Hence, these checks can only be performed, if an assignment table is present.

4.3.1 Requirements for an assignment table

An assignment table with unique assignments between ICAO 3-letter country codes and International Organization for Standardization (ISO) 2-letter country codes SHALL be present according to [ISO3166-2].

This assignment table SHALL be prepared in coordination between the operator and vendor according to [ISO3166-1] and [ISO3166-2]. Here, the values according to [ISO3166-1] ALPHA-2 and ALPHA-3 codes as well as any deviating two-letter codes of countries with territorial divisions according to [ISO3166-2] are taken

into consideration. The changes and other adjustments published in the [ISO3166-1] newsletter SHALL be incorporated into maintenance, in coordination with the operator and vendor.

This assignment is not necessarily unambiguous for each country; for example, an ISO 2-letter country code can be assigned to several ICAO 3-letter country codes and, in turn, an ICAO 3-letter country code can also be assigned to several ISO 2-letter country codes. Such cases SHALL be discussed between the operator and vendor, and SHOULD be specified in the corresponding profiling.

5 Authentication of MRTDs

5.1 Process sequence of checking an MRTD

The following process steps are usually performed when checking MRTDs:

- Reading the document optically
- Checking the recorded features
- Reading the document electronically
- Checking the electronic features
- Performing combined checks

It is RECOMMENDED to follow the process sequence shown in ►Figure 5.1:

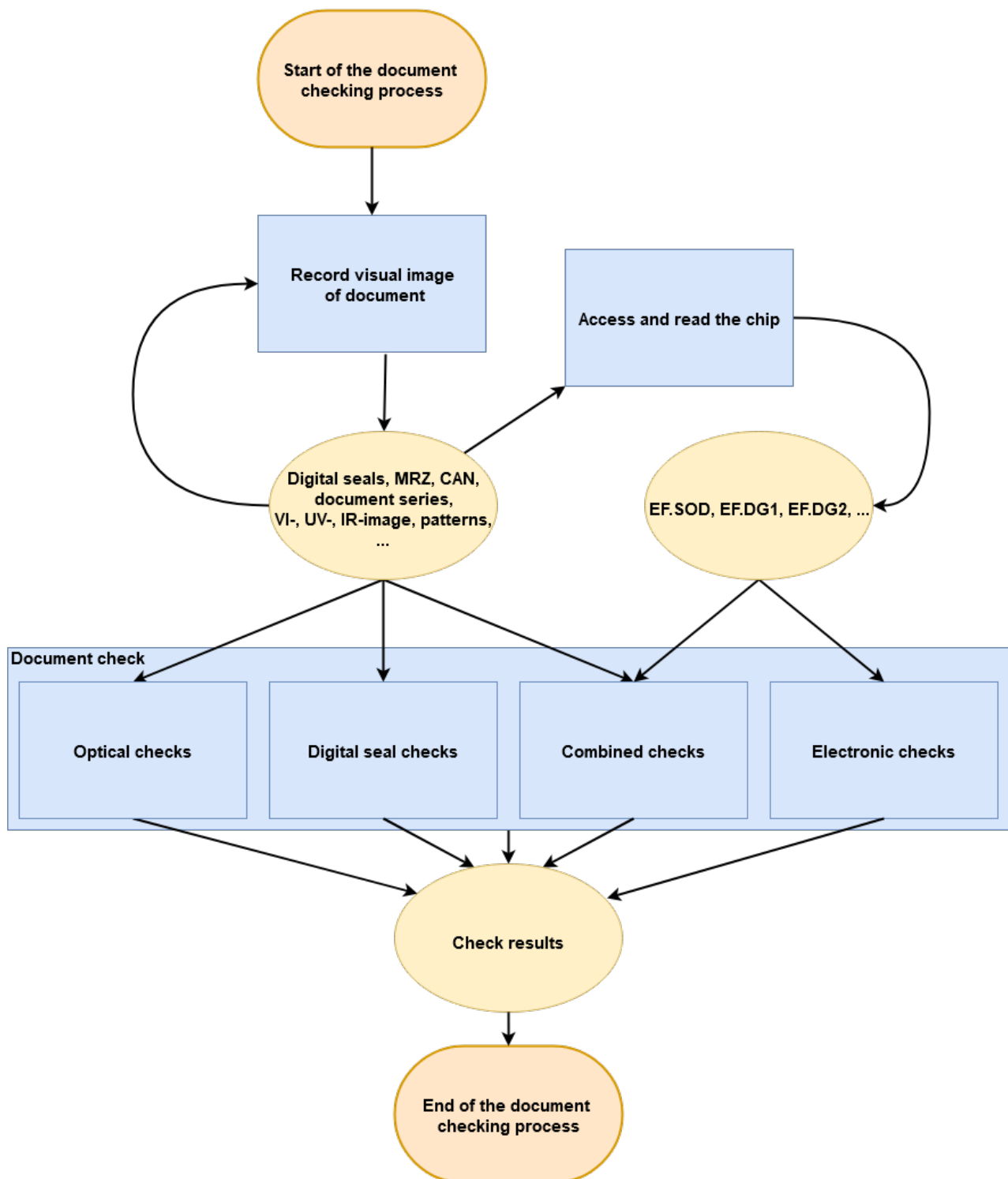


Figure 5.1. Process sequence of checking MRTDs

The process starts with reading the optical/physical features of an MRTD. Depending on the application scenario, the optical images of the data page and the MRZ or the CAN is read ¹. Based on this data, a logical check (e. g. checking the validity of the document) as well as a check of the optical/physical security features (including digital seals) of the document are performed. ▶ Section 5.7 specifies the optical checks on MRTDs. ▶ Section 5.9 specifies the digital seal checks on MRTDs.

¹ Depending on the document, several pages (e. g. data page and visa page) can be checked and assorted to one result. Checking several documents within one transaction is considered as to be several individual checks.

Reading the document electronically can be started as soon as the MRZ or the CAN of the document is available and a RF chip is identified. The architecture and procedure required for reading eMRTDs electronically is specified in ▶Section 5.8.

Combined checks SHOULD be performed as soon as the data required for such checks are available, see ▶Section 5.10.

5.2 Visualization of check results

Inspection Applications SHALL be able to provide the results of an individual check with reduced level of detail. It is RECOMMENDED to map the defined values as defined in ▶Section 3.6.2 following a scheme based on traffic light colours or pictograms as shown in ▶Table 5.1. The visualization schemes based on traffic light colours and pictograms MAY be combined. Other geometric shapes MAY be used in pictograms instead of circles.

Result of the check	Traffic light colour	Pictogram description	Pictogram	Traffic light with pictogram
successful	green	check mark in a circle		
failed	red	cross mark in a circle		
undetermined	yellow	exclamation mark in a circle		
not supported	grey	empty circle		
aborted	black	stop mark in a circle		

Table 5.1 Assigning the results of checks using traffic light colours and pictogram symbolism

5.3 Error codes

Each of the checks prescribed in ▶Section 5.7, ▶Section 5.8, ▶Section 5.9 and ▶Section 5.10 SHOULD log processing errors using numeric integer error codes. Each error SHALL distinguish between the class of a check (i. e. Optical checks, Electronic checks, Digital Seal checks and Combined checks).

For each defined error code in each class, the Inspection Application SHOULD also provide a short and concise error description that facilitates a subsequent analysis regarding the error. Examples of errors are program crashes, improper use and hardware failures. Errors may result in an abortion of the defined checking process.

Vendor-specific error codes SHALL be defined in the integer range of 1.000 – 999.999, error codes beyond that range are reserved for use by this Technical Guideline and are subject to be defined in the profiling document.

No error code SHALL mean that no error occurred and that a process was performed successfully. The error code value can be set to a NULL or VOID value.

5.4 Overall document check result

Checks from ▶Section 5.7, ▶Section 5.8, ▶Section 5.9 and ▶Section 5.10 SHALL be combined into an overall document check result. The aggregation into a single result is Inspection Application specific. The overall document result check SHALL be mapped according to ▶Table 5.2.

Result of the check	Description
successful	The check result SHALL be set to <code>successful</code> if all assorted overall higher-level checks from ▶Section 5.7, ▶Section 5.8, ▶Section 5.9 and ▶Section 5.10 are <code>successful</code> .
failed	The check result SHALL be set to <code>failed</code> if at least one of the assorted higher-level checks from ▶Section 5.7, ▶Section 5.8, ▶Section 5.9 and ▶Section 5.10 was set to <code>failed</code> .
undetermined	The check result can be <code>undetermined</code> depending on an Inspection Application's specific assessment.

Result of the check	Description
not supported	The check result can be not supported depending on an Inspection Application's specific assessment.
aborted	The check result SHALL be set to aborted if at least one of the assorted higher-level checks from ▶Section 5.7, ▶Section 5.8, ▶Section 5.9 and ▶Section 5.10 was aborted.

Table 5.2 Overall Document Check Result

5.5 Aggregation of overall check results

The overall result of a document check is an aggregation of the overall results of the optical, electronic, combined, and digital seal checks. These various checks are available to ensure the authenticity, integrity, confidentiality and availability of MRTDs and are described in detail in the following sections. The aggregation of these check results is summarised in ▶Table 5.3.

Document Check	Partial Checks
Overall Check	▶Section 5.7.1: Aggregation of optical check results
Checks	OpticalChecks
	▶Section 5.8.1: Aggregation of electronic check results
	ElectronicChecks
	▶Section 5.9.1.1: Aggregation of digital seal check results
	SealChecks
	▶Section 5.10.1: Aggregation of combined check results
	CombinedCheck

Table 5.3 Aggregation of overall check results

5.6 Common checks

This section describes checks that are relevant for checking several different types of document components. These common checks are referenced in the corresponding sections that describe the checks for the different document components.

5.6.1 Check the MRZ data

In the following sections, MRZs data checks are explained in detail. The checks that SHALL be performed depend on the application scenario and the individual document. They SHOULD be defined in a profiling document.

A document can encompass various components (printed MRZ, EF.DG1, digital seal) that contain MRZ data. The IS SHALL check the MRZs data in each document component.

This section list checks that can be applied to MRZ data from any component. Additional checks specific to a printed MRZ are given in ▶Section 5.7: Checking an optical component.

5.6.1.1 Aggregation of MRZ data consistency check results

The MRZ data consistency check consists of several partial checks that are aggregated together. The previously specified aggregation rules from ▶Section 3.8 are applied to all partial check results. The aggregation of these check results is shown in ▶Table 5.4 below.

Check	Partial Checks
▶Section 5.6.1.2: Check the consistency of the MRZ data	▶Section 5.6.1.2.1: Check the length of the MRZ data
MrzConsistency-Check	LengthCheck
	▶Section 5.6.1.2.2: Check the composite check digit
	CheckDigitCompositeCheck

Check	Partial Checks
	▶Section 5.6.1.2.3: Check the document number check digit CheckDigitDocumentNumberCheck
	▶Section 5.6.1.2.4: Check the date of expiry check digit CheckDigitDateOfExpiryCheck
	▶Section 5.6.1.2.5: Check the date of birth check digit CheckDigitDateOfBirthCheck
	▶Section 5.6.1.2.6: Check the optional data check digit CheckDigitOptionalDataCheck
	▶Section 5.6.1.2.7: Check the consistency between issuer and nationality IssuerVsNationalityCheck
	▶Section 5.6.1.2.8: Check the document type DocumentTypeCheck
	▶Section 5.6.1.2.9: Check the issuer IssuerCheck
	▶Section 5.6.1.2.10: Check the nationality NationalityCheck
	▶Section 5.6.1.2.11: Check the date of birth DateOfBirthCheck
	▶Section 5.6.1.2.12: Check the date of expiry DateOfExpiryCheck
	▶Section 5.6.1.2.13: Check the sex of the MRZ SexCheck
	▶Section 5.6.1.2.14: Check the document number DocumentNumberCheck
	▶Section 5.6.1.2.15: Check the surname SurnameCheck
	▶Section 5.6.1.2.16: Check the given name GivenNameCheck
	▶Section 5.6.1.2.17: Check whether the issuer is included in the ICAO list IssuerIcaoIncludedCheck
	▶Section 5.6.1.2.18: Check whether the nationality is included in the ICAO list NationalityIcaoIncludedCheck

Table 5.4 Details of a consistency check of the MRZ data

5.6.1.2 Check the consistency of the MRZ data

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario. ICAO conformity checks SHALL use [ICAO9303].

The aggregation result SHALL be mapped according to ▶Table 5.5.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.

Result	Description
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.5 Result of checking the consistency

5.6.1.2.1 Check the length of the MRZ data

The IS SHALL check if length (number of characters) of the MRZ data is ICAO compliant.

The result of this check SHALL be mapped according to ►Table 5.6.

Result	Description
successful	The length of the MRZ data is ICAO compliant.
failed	The length of the MRZ data is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	The document type does not contain MRZ data.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.6 Result of checking the length

5.6.1.2.2 Check the composite check digit

The IS SHALL check the correctness of the composite check digit.

The result of this check SHALL be mapped according to ►Table 5.7.

Result	Description
successful	The composite check digit is correct.
failed	The composite check digit is incorrect or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.7 Result of checking the composite check digit

5.6.1.2.3 Check the document number check digit

The IS SHALL check the correctness of the document number check digit.

The result of this check SHALL be mapped according to ►Table 5.8.

Result	Description
successful	The document number check digit is correct.
failed	The document number check digit is incorrect or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.

Result	Description
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.8 Result of checking the document number check digit

5.6.1.2.4 Check the date of expiry check digit

The IS SHALL check the correctness of the date of expiry check digit.

The result of this check SHALL be mapped according to ►Table 5.9.

Result	Description
successful	The date of expiry check digit is correct.
failed	The date of expiry check digit is incorrect or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.9 Result of checking the date of expiry check digit

5.6.1.2.5 Check the date of birth check digit

The IS SHALL check the correctness of the date of birth check digit.

The result of this check SHALL be mapped according to ►Table 5.10.

Result	Description
successful	The date of birth check digit is correct.
failed	The date of birth check digit is incorrect or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.10 Result of checking the date of birth check digit

5.6.1.2.6 Check the optional data check digit

The IS SHALL check the correctness of the optional check digit.

The result of this check SHALL be mapped according to ►Table 5.11.

Result	Description
successful	The optional data check digit is correct.
failed	The optional data check digit is incorrect or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.11 Result of checking the optional data check digit

5.6.1.2.7 Check the consistency between issuer and nationality

The IS SHALL check whether the issuer and the nationality are consistent with each other.

The result of this check SHALL be mapped according to ►Table 5.12.

Notes:

1. The logic of this check depends on the document type (e.g. visa vs. passport).
2. Even for passports it is not sufficient to check for equality because certain countries may issue passports for non nationals or may differentiate between citizenship and nationality.

Result	Description
successful	The issuer and nationality are consistent with each other.
failed	The issuer and nationality are not consistent with each other or at least one of them is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	At least one of the two comparands is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.12 Result of checking the consistency between issuer and nationality

5.6.1.2.8 Check the document type

The IS SHALL check whether the document type is ICAO compliant.

The result of this check SHALL be mapped according to ►Table 5.13.

Result	Description
successful	The document type is ICAO compliant.
failed	The document type is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.13 Result of checking the document type

5.6.1.2.9 Check the issuer

The IS SHALL check whether the issuer is ICAO compliant. Whether the issuer is included in the ICAO list of known nationalities (i.e. country codes) SHALL NOT be part of this check.

The result of this check SHALL be mapped according to ►Table 5.14.

Result	Description
successful	The issuer is ICAO compliant.
failed	The issuer is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.14 Result of checking the issuer

5.6.1.2.10 Check the nationality

The IS SHALL check whether the nationality is ICAO compliant. Whether the nationality is included in the ICAO list of known nationalities (i.e. country codes) SHALL NOT be part of this check.

The result of this check SHALL be mapped according to ►Table 5.15.

Result	Description
successful	The nationality is ICAO compliant.

Result	Description
failed	The nationality is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.15 Result of checking the nationality

5.6.1.2.11 Check the date of birth

The IS SHALL check whether the date of birth is ICAO compliant. Note that a year of birth consists of the century digits (CC) and the remaining digits (YY), i.e. CCYY. For example, for 1989 we derive that CC is equal to 19 and YY is equal to 89.

If the year of birth is derived from a two digit year number (YY), e.g. from an (optical) MRZ, the two century digits (CC) SHALL be determined as follows:

Let IIKK be the year of the date of inspection, i.e. II are the century digits of the date of inspection. If IIYY is greater than or equal to the year of the date of inspection IIKK, the year of birth is set to (II - 1)YY. Otherwise, the year of birth is set to IIYY.

The result of this check SHALL be mapped according to ► Table 5.16.

Result	Description
successful	The date of birth is ICAO compliant.
failed	The date of birth is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.16 Result of checking the date of birth

5.6.1.2.12 Check the date of expiry

The IS SHALL check whether the date of expiry is ICAO compliant.

The result of this check SHALL be mapped according to ► Table 5.17.

Result	Description
successful	The date of expiry is ICAO compliant.
failed	The date of expiry is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.17 Result of checking the date of expiry

5.6.1.2.13 Check the sex of the MRZ

The IS SHALL check whether the sex is ICAO compliant.

The result of this check SHALL be mapped according to ► Table 5.18.

Result	Description
successful	The sex is ICAO compliant.

Result	Description
failed	The sex is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.18 Result of checking the sex

5.6.1.2.14 Check the document number

The IS SHALL check whether the document number is ICAO compliant.

The result of this check SHALL be mapped according to ► Table 5.19.

Result	Description
successful	The document number is ICAO compliant.
failed	The document number is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.19 Result of checking the document number

5.6.1.2.15 Check the surname

The IS SHALL check whether the surname is ICAO compliant.

The result of this check SHALL be mapped according to ► Table 5.20.

Result	Description
successful	The surname is ICAO compliant.
failed	The surname is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.20 Result of checking the surname

5.6.1.2.16 Check the given name

The IS SHALL check whether the given name is ICAO compliant.

The result of this check SHALL be mapped according to ► Table 5.21.

Result	Description
successful	The given name is ICAO compliant.
failed	The given name is not ICAO compliant or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.21 Result of checking the given name

5.6.1.2.17 Check whether the issuer is included in the ICAO list

The IS SHALL check whether the issuer is included in the ICAO list of known issuers (i.e. country codes).

The result of this check SHALL be mapped according to ►Table 5.22.

Result	Description
successful	The issuer is included in the ICAO list.
failed	The issuer is not included in the ICAO list or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.22 Result of checking whether the issuer is included in the ICAO list

5.6.1.2.18 Check whether the nationality is included in the ICAO list

The IS SHALL check whether the nationality is included in the ICAO list of known nationalities (i.e. country codes).

The result of this check SHALL be mapped according to ►Table 5.23.

Result	Description
successful	The nationality is included in the ICAO list.
failed	The nationality is not included in the ICAO list or this datum is expected to be present for the identified document model but is not present.
undetermined	This result SHALL NOT occur here.
not supported	This datum is not expected to be present for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.23 Result of checking whether the nationality is included in the ICAO list

5.6.2 Issuing country checks

In the following sections, checks regarding the origin of certificates are listed. The checks to perform depend on the application scenario and the individual document models. These SHOULD be defined in a profiling document.

The certificates contain information about the issuer and signer that SHALL be checked for consistency. These checks use the certificates but also MRZs data contained in the document component that is checked.

The ISs SHALL check the certificates in each document component and log the results.

5.6.2.1 Check of issuing country information in certificates

There are several checks to perform checks about the issuing country or authority of certificates. An overview of these checks is given in ►Table 5.24 .

Checks
►Section 5.6.2.2.1: Check the consistency of issuer DN and MRZ data SignerIssuerDnVsMrzDataCheck
►Section 5.6.2.2.2: Check the consistency of subject DN and MRZ SignerSubjectDnVsMrzDataCheck
►Section 5.6.2.2.3: Check the consistency of subject DN and issuer DN SignerSubjectDnVsIssuerDnCheck

Table 5.24 Details of an issuing country check

5.6.2.2 Check a signer certificate issuing country or organization

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario. This check and its partial checks SHALL be performed separately for every signer certificate used in the check.

The IS SHALL fulfill the requirements in ▶Section 4.3.1 for the partial checks.

The aggregation result SHALL be mapped according to ▶Table 5.25

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.25 Result of checking a signer certificate

5.6.2.2.1 Check the consistency of issuer DN and MRZ data

The IS SHALL check whether the issuing country in the Distinguished Name (DN) and MRZ data are consistent, i.e. whether the issuing country or organization (C=) in the signer certificate issuer distinguished name corresponds to the MRZ data present in EF.DG1 or digital seal, respectively.

The result of this check SHALL be mapped according to ▶Table 5.26.

Result of the check	Description
successful	The country entry in the issuer DN and the issuer country code in the MRZ data are consistent.
failed	The country entry in the issuer DN and the issuer country code in the MRZ data are not consistent, or at least one of the components is missing but was expected for the identified document model.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.26 Result of checking the consistency of issuer DN and MRZ data

5.6.2.2.2 Check the consistency of subject DN and MRZ

The IS SHALL check whether the issuing country in the DN and MRZ are consistent, i.e. whether the issuing country or organization (C=) in the signer certificate subject distinguished name corresponds to the MRZ data present in EF.DG1 or digital seal, respectively.

The result of this check SHALL be mapped according to ▶Table 5.27.

Result of the check	Description
successful	The country entry in the subject DN and the issuer country code in the MRZ data are consistent.

Result of the check	Description
failed	The country entry in the subject DN and the issuer country code in the MRZ data are not consistent or at least one of the components is missing but was expected for the identified document model.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.27 Result of checking the consistency of subject DN and MRZ

5.6.2.2.3 Check the consistency of subject DN and issuer DN

The IS SHALL check whether the issuing country in subject DN and issuer DN are consistent, i.e. whether the issuing country or organization (C=) in the signer certificate are identical for subject and issuer.

The result of this check SHALL be mapped according to ►Table 5.28.

Result of the check	Description
successful	The country entries in the issuer and subject DNs match.
failed	The country entries in the issuer and subject DNs do not match.
undetermined	This result SHALL NOT occur here.
not supported	The document does not contain a certificate.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.28 Result of checking the consistency of issuer DN and issuer DN

5.6.3 Check of the certificates

In the following sections, certificate checks are explained in detail. The checks that SHALL be performed depend on the application scenario and the individual document. These SHOULD be defined in a profiling document.

A document can encompass various components (chip, DTCs, digital seals) that contain certificates.

The ISs SHALL check the certificates in each document component and log the results.

5.6.3.1 Aggregation of certificate check results

The certificate check consists of several partial checks that are aggregated together. The previously specified aggregation rules from ►Section 3.8 are applied to all partial check results. The aggregation of these check results is shown in ►Table 5.29 below.

Check	Partial Checks
►Section 5.6.3.2: Check the issuer certificates of a document component IssuerCertificateCheck SecurityFileCheck	►Section 5.6.3.2.1: Check the signer certificate's signature SignerSignatureCheck
	►Section 5.6.3.2.2: Check the signer certificate's validity period SignerValidityPeriodCheck
	►Section 5.6.3.2.3: Check the country signer certificate's validity period CountrySignerValidityPeriodCheck
	►Section 5.6.3.2.4: Check the signer certificate's trust status SignerTrustedCheck
	►Section 5.6.3.2.5: Check that the signer certificate is not self-signed SignerNotSelfSignedCheck
	►Section 5.6.3.2.6: Check the signer certificate's subject distinguished name SignerSubjectDnCheck

Check	Partial Checks
	►Section 5.6.3.2.7: Check the signer certificate's issuer distinguished name SignerIssuerDnCheck

Table 5.29 Details of a issuer certificate check

5.6.3.2 Check the issuer certificates of a document component

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The data contained inside a document component is secured by digital signatures. To verify a digital signature the IS needs the corresponding public key. These public keys are contained inside certificates. Each certificate has its own digital signature which SHALL be validated with the public key of another certificate except for the certificate at the root of the chain of certificates. The certificate at the root of the chain of certificates is signed by itself.

In the context of this TR the chain of certificates has two levels: The CSCA certificate at the root of the chain and a signer certificate (e.g. DS certificate) below it.

The signer certificates are either stored directly in the document component (e.g. EF.SOD, EF.CardSecurity, EF.ChipSecurity for an electronic component) or contained in a Signer List. The CSCA certificate always has to be provided from an external source. This is usually the Master List. Master Lists and the Signer Lists are published by the national PKD service and provided via the IS's background infrastructure (see [BSI TR-03129-2]) in a trusted procedure.

In order to verify the digital signature of the signer certificate, the IS SHALL acquire the corresponding CSCA certificate. The corresponding CSCA certificate could be searched for by the following mechanisms:

- It is RECOMMENDED that an IS extracts the `AuthorityKeyIdentifier` extension from the signer certificate and thus searches for the actual CSCA certificate using the corresponding `SubjectKeyIdentifier` extension. The country codes of both certificates or their locality names SHALL be cross-checked.
- Although this extension is mandatory according to [ICAO9303], there are countries that issue signer certificates without a `AuthorityKeyIdentifier` or/and CSCA certificates without a `SubjectKeyIdentifier`. For these special cases, and only for these, the following is RECOMMENDED:

If no corresponding CSCA certificate can be found via the certificate extensions, the IS can perform a search based on the issuer information (Issuer) of the signer certificate. If more than one corresponding CSCA certificate is found, checking the signer certificate's digital signature (see ►Section 5.6.3.2.1) SHALL be set to `successful` if checking the digital signature using one of the CSCA certificates found was successful and the owner (Subject) of the CSCA certificate corresponds to the issuer (Issuer) of the signer certificate. When performing the search via the issuer name, `failed` can thus not occur as result.

- If there is only a CSCA link certificate, the digital signature of this CSCA link certificate SHALL NOT be verified, since the check would fail.

The aggregation result SHALL be mapped according to ►Table 5.30.

Result	Description
<code>successful</code>	The result SHALL be set to <code>successful</code> if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result <code>failed</code> , <code>aborted</code> or <code>undetermined</code> and at least one had the result <code>successful</code> .
<code>failed</code>	The result SHALL be set to <code>failed</code> if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result <code>failed</code> .
<code>undetermined</code>	The result SHALL be set to <code>undetermined</code> if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result <code>failed</code> or <code>aborted</code> and at least one had the result <code>undetermined</code> .
<code>not supported</code>	The result SHALL be set to <code>not supported</code> if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result <code>not supported</code> .

Result	Description
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.30 Result of checking the issuer certificates of a document component

5.6.3.2.1 Check the signer certificate's signature

The IS SHALL verify the signer certificate's signature.

The result of this check SHALL be mapped according to ►Table 5.31.

Result	Description
successful	The signer certificate's signature is valid.
failed	The signer certificate's signature is not valid.
undetermined	Information is missing for verifying the signature (e.g. the signature algorithm is unknown or unavailability of a CSCA certificate or the signer certificate is unparsable or not available).
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.31 Result of checking the signer's signature

If the defect `Wrong Signer Identifier` applies, the requirements specified in section ►Section 5.12.2.2.4 SHALL be implemented.

If the defect `Document Signer Certificate incorrectly encoded or malformed` applies, the requirements specified in ►Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.2 Check the signer certificate's validity period

The IS SHALL check whether the signer certificate is valid at the time of the check. Timestamps used for this check SHALL be provided by a reliable time service.

The result of this check SHALL be mapped according to ►Table 5.32.

Result	Description
successful	The time of inspection is inside the validity period.
failed	The time of inspection is not inside the validity period.
undetermined	The IS cannot determine the time of inspection, cannot acquire a validity period or the signer certificate is unparsable or unavailable.
not supported	This result can only occur if there is a known defect.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.32 Result of checking the signer's validity period

If the defect `Document Signer Certificate Malformed` is included in the Defect List and can be identified, the requirements specified in ►Section 5.12.2.1.3 SHALL be implemented.

If the defect `Validity Period Incorrect` applies with parameters `DS_validity` or `CSCA_DS_validity` (see ►Section 5.12.2.1.7) the result SHALL be not supported.

If the defect `Document Signer Certificate incorrectly encoded or malformed` applies, the requirements specified in ►Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.3 Check the country signer certificate's validity period

The IS SHALL check whether the country signer certificate is valid at the time of the check. Timestamps used for this check SHALL be provided by a reliable time service.

The result of this check SHALL be mapped according to ►Table 5.33.

Result	Description
successful	The time of inspection is inside the validity period.
failed	The time of inspection is not inside the validity period.
undetermined	The IS cannot determine the time of inspection, cannot acquire a validity period or the country signer certificate is not available.
not supported	This result can only occur if there is a known defect.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.33 Result of checking the country signer's validity period

If the defect `Validity Period Incorrect` applies with `CSCA_validity` or `CSCA_DS_validity` (see ▶Section 5.12.2.1.7) the result SHALL be not supported.

If the defect `Document Signer Certificate incorrectly encoded or malformed` applies, the requirements specified in ▶Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.4 Check the signer certificate's trust status

The IS SHALL check the trust status of the signer certificate. The basis of this check are the Master List, Signer List, Defect List and Deviation List.

The result of this check SHALL be mapped according to ▶Table 5.34.

Result	Description
successful	The IS trusts the signer certificate.
failed	The IS does not trust the signer certificate.
undetermined	The IS cannot assess the trust status because it lacks necessary information to perform the assessment (e.g. the certificate is unparsable or unavailable).
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.34 Result of checking the signer's trust status

If the defect `Document Signer Certificate incorrectly encoded or malformed` applies, the requirements specified in ▶Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.5 Check that the signer certificate is not self-signed

The IS SHALL check whether the signer certificate is not self-signed, i.e. its public key does NOT validate its own signature.

The result of this check SHALL be mapped according to ▶Table 5.35.

Result	Description
successful	The signer certificate is not self-signed.
failed	The signer certificate is self-signed.
undetermined	Information is missing to perform the check (e.g. an unsupported algorithm is used) or the certificate is unparsable or not available.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.35 Result of checking that the signer is not self-signed

If the defect `Document Signer Certificate incorrectly encoded or malformed` applies, the requirements specified in ▶Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.6 Check the signer certificate's subject distinguished name

The IS SHALL check the subject distinguished name of the signer certificate.

The result of this check SHALL be mapped according to ▶Table 5.36.

Result	Description
successful	The fields <code>countryName</code> and <code>commonName</code> are present and the values are syntactically correct.
failed	One or both of the fields <code>countryName</code> and <code>commonName</code> are not present or the corresponding values are syntactically incorrect.
undetermined	The certificate is not parsable or unavailable.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.36 Result of checking the signer's subject distinguished name

If the defect Document Signer Certificate incorrectly encoded or malformed applies, the requirements specified in ▶Section 5.12.2.3.4 SHALL be implemented.

5.6.3.2.7 Check the signer certificate's issuer distinguished name

The IS SHALL check the issuer distinguished name of the signer certificate.

The result of this check SHALL be mapped according to ▶Table 5.37.

Result	Description
successful	The fields <code>countryName</code> and <code>commonName</code> are present and the values are syntactically correct.
failed	One or both of the fields <code>countryName</code> and <code>commonName</code> are not present or the corresponding values are syntactically incorrect.
undetermined	The certificate is not parsable or unavailable.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.37 Result of checking the signer's issuer distinguished name

If the defect Document Signer Certificate incorrectly encoded or malformed applies, the requirements specified in ▶Section 5.12.2.3.4 SHALL be implemented.

5.6.4 Check the consistent representation of one component

The IS SHALL check whether two given comparands, which represent the same component (e.g. document number, surname etc.) but originate from different sources (optical MRZ, EF.DG1, Visual Zone (VIZ) or digital seal), are consistent with each other.

The result of this check SHALL be mapped according to ▶Table 5.38.

Result	Description
successful	The two comparands are consistent with each other.
failed	The two comparands are not consistent with each other or one comparand is missing but expected to be present for the identified document model.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.38 Result of checking the consistency between two comparands

5.7 Checking an optical component

The following sections describe how an IS SHALL inspect an optical component.

The specific checks to be performed depend on the application scenario and the individual document models. They SHOULD be defined in a profiling document.

5.7.1 Aggregation of optical check results

The optical check consists of several partial checks that are aggregated together. The previously specified aggregation rules from ▶Section 3.8 are applied to all partial check results. The associated return value is translated into traffic light colors according to ▶Table 5.1 and displayed as the result of the optical check. The aggregation of these check results is shown in ▶Table 5.39 below. Each row describes an optical check, which is divided into partial checks. A further subdivision of the check is described after each further column to the right.

Document Check	Partial Checks	Partial Checks	Partial Checks
Optical Check Checks	▶Section 5.7.3.1: Document detection and capturing DocumentDetectionCheck		
	▶Section 5.7.3.2: Document model identification DocumentModelIdentificationCheck		
	▶Section 5.7.3.3: Check the document for outdated model NoOutdatedModelCheck		
	▶Section 5.7.3.4: Check for test document NoTestDocumentCheck		
	▶Section 5.7.3.5: Check if the document is only valid in the issuing country NoDomesticDocumentCheck		
	▶Section 5.7.3.6: Check for acknowledged issuer AcknowledgedIssuerCheck		
	▶Section 5.7.3.7: Check for the presence of the primary facial image PrimaryFacialImagePresenceCheck		
	▶Section 5.7.3.8: Check for the embedding of the primary facial image PrimaryFacialImageEmbeddingCheck		
	▶Section 5.7.3.9: Check the ICAO compliance of MRZ MrzComplianceCheck	▶Section 5.7.3.9.1: Check the consistency of MRZ data MrzConsistencyCheck	Partial checks see ▶Table 5.4.
		▶Section 5.7.3.9.2: Check MRZ to be readable under IR light MrzIrReadabilityCheck	
		▶Section 5.7.3.9.3: Check MRZ to be printed using OCR-B type face MrzOcrbTypeFaceCheck	

Document Check	Partial Checks	Partial Checks	Partial Checks
		▶Section 5.7.3.9.4: Check alignment of MRZ MrzAlignmentCheck	▶Section 5.7.3.9.4.1: Check character alignment of MRZ CharacterAlignmentCheck
			▶Section 5.7.3.9.4.2: Check horizontal alignment of MRZ HorizontalAlignmentCheck
			▶Section 5.7.3.9.4.3: Check vertical alignment of MRZ VerticalAlignmentCheck
	▶Section 5.7.3.10: VIZ consistency check VisualZoneConsistencyCheck	Partial checks see ▶Table 5.40: Details of a consistency check of the VIZ.	
	▶Section 5.7.3.11: Check for the consistency of MRZ and VIZ MrzVsVizCheck	▶Section 5.6.4: Check the consistent representation of one component ComponentCheck	
	▶Section 5.7.3.12.3: Perform Spectrally Selective Verification SpectrallySelectiveVerificationCheck	▶Section 5.7.3.12.3.1: Check spectrally selective verification group check SsvGroupCheck	▶Section 5.7.3.12.3.1: Check basic spectrally selective checks BasicCheck
			▶Section 5.7.3.12.3.1: Check composite spectrally selective checks CompositeCheck

Table 5.39 Aggregation of optical document check results

5.7.1.1 Check for the consistency of VIZ

The IS SHALL check if the VIZ is in compliance with ICAO specifications. The details of these consistency checks are shown in ▶Table 5.40.

Check	Partial Checks
▶Section 5.7.1.1: Check for the consistency of VIZ. VisualZoneConsistencyCheck	▶Section 5.7.3.10.1: VIZ's issuer versus VIZ's nationality consistency check IssuerVsNationalityCheck
	▶Section 5.7.3.10.2: VIZ's document type consistency check DocumentTypeCheck
	▶Section 5.7.3.10.3: VIZ's issuer consistency check IssuerCheck
	▶Section 5.7.3.10.4: VIZ's nationality consistency check NationalityCheck
	▶Section 5.7.3.10.5: VIZ's date of birth consistency check DateOfBirthCheck
	▶Section 5.7.3.10.6: VIZ's date of expiry consistency check DateOfExpiryCheck
	▶Section 5.7.3.10.7: VIZ's sex consistency check SexCheck
	▶Section 5.7.3.10.8: VIZ's document number consistency check DocumentNumberCheck

Check	Partial Checks
	▶Section 5.7.3.10.9: VIZ's surname consistency check SurnameCheck
	▶Section 5.7.3.10.10: VIZ's given name consistency check GivenNameCheck
	▶Section 5.7.3.10.11: VIZ's issuer ICAO included check IssuerIcaoIncludedCheck
	▶Section 5.7.3.10.12: VIZ's nationality ICAO included check NationalityIcaoIncludedCheck
	▶Section 5.7.3.10.13: VIZ's authority or issuing organization consistency check AuthorityOrIssuingOrganizationCheck
	▶Section 5.7.3.10.14: VIZ's Card Access Number (CAN) consistency check CardAccessNumberCheck
	▶Section 5.7.3.10.15: VIZ's date of issue consistency check DateOfIssueCheck
	▶Section 5.7.3.10.16: VIZ's place of issue consistency check PlaceOfIssueCheck

Table 5.40 Details of a consistency check of the VIZ

5.7.2 Introduction to spectrally selective checks

In this section, a base set of optical checks is described which is required for further checks. The main focus of these checks are spectrally selective check routines.

The sequence of optical checks is given in ▶Figure 5.2. Regarding this sequence the model identification (see ▶Section 5.7.2.1 and ▶Section 5.7.3.2) and checking the MRZ consistency (see ▶Section 5.7.3.9) SHALL be performed for all documents, whereas most of the spectrally selective checks can only be performed after the document model was identified successfully.

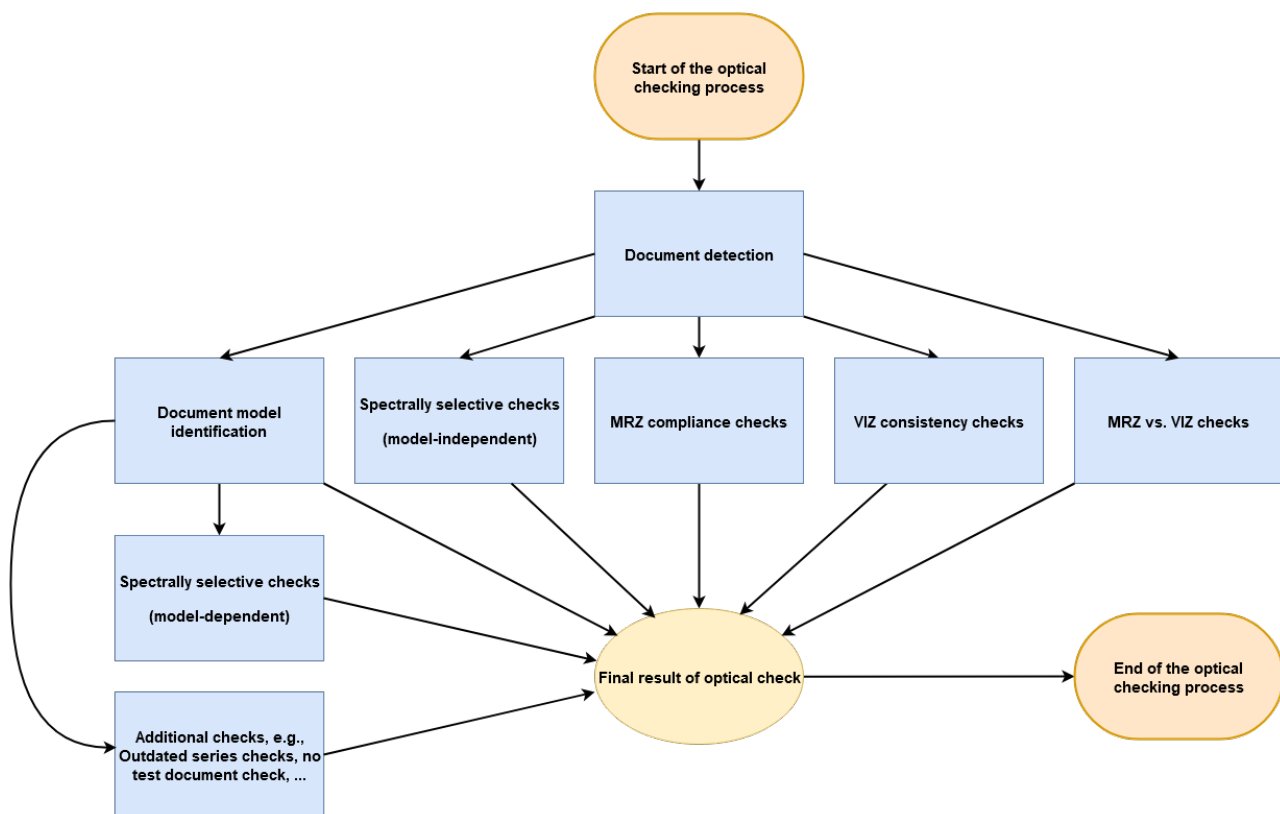


Figure 5.2. Example sequence of optical checks of MRTDs

5.7.2.1 Document model

Each MRTD can be assigned to a document model (sometimes also called document series). A document model is defined by means of the country code (**C**), document type (**T**), a distinct identification number (**N**), a year value (**Y**), a PRADO [PRADO] code identifier (**P**), a subdivision or state (**S**) and an organization (**O**) :

Document Model := (**C**, **T**, **N**, **Y**, **P**, **S**, **O**)

The country code **C** SHALL be a three-letter code according to [ICAO9303].

The first digit of the document type **T** is specified in [ICAO9303].

The identification number **N** SHALL be a distinct chronological increasing integer number starting with 1.

The year **Y** refers to the year as an integer value if the year is unknown this value SHALL be omitted.

The value for **P** represents the PRADO code identifier. If there is no known PRADO code, this value SHALL be omitted.

The value for **S** represents the subdivision or state. If there is no known subdivision or state, this value SHALL be omitted.

The value for **O** represents the organisation. If there is no known organisation, this value SHALL be omitted.

5.7.2.2 Spectrally selective checks

The term "spectrally selective checks" covers checks, that record different reactions occurring on a document illuminated with visual (**VI**, white light) or non-visual (**UV**, **IR**) illumination. Based on the three records (**VI**, **UV**, **IR**), the absorbent, reflective or luminescent features of these reactions can be checked. Examples for checks are the MRZ IR readability, the VIZ IR readability, the UV brightness test and sample verifications in all images. With a few exceptions these checks can only be defined for a specific model of documents. Hence a check database is required to assist the checking process. Such a database contains model-specific spectrally selective check information about known documents. The actual characteristics of a spectrally selective check routine depend on the vendor of the inspection system.

For vendor-independent comparability check results, generic designations are necessary and will be specified in the following paragraphs. Basic check routines are individual check routines which refer to an individual feature (for example, the IR absorption). Composite check routines are defined as logical combinations of basic check routines. Thus, an enhanced characteristic can be checked with respect to several features, such as IR absorption and visibility in the VI.

For the check routines, the following abbreviated definitions are used:

(XX, YY, ZZ)

XX specifies the light source of the record on which the check routine is performed:

- **IR** - infrared light
- **UV** - ultraviolet light
- **VI** - visible (white) light

YY specifies the optical property of the characteristic:

- **AB** - absorbent
- **BR** - bright
- **FR** - spatial frequency property
- **LU** - luminescent
- **TL** - translucent
- **TR** - transparent

ZZ specifies the characteristic feature of or the position in the document ²:

- **FI** - fibres
- **FU** - full (complete) data page
- **IS** - printed feature, which already exists on the substrate (Ink static)
- **MR** - area of the machine readable zone (MRZ)
- **OM** - overprinted MRZ
- **CA** - Card Access Number (CAN)
- **PD** - personalized "dynamic" perforation
- **PS** - perforation which shows "static" content
- **PH** - area of the photo
- **SP** - area of the secondary photo
- **OP** - overprinted photo
- **TH** - security thread
- **VZ** - visual inspection zone (VIZ)
- **WM** - watermark
- **ID** - any other personalized, "dynamic" feature (ink dynamic), e. g. a secondary photograph
- **AF** - any additional feature that cannot be attributed to the items specified above

The vendor SHALL assign a unique identifier (Universally Unique Identifier (UUID) according to [RFC4122]) for each generic check. For example, (**IR**, **AB**, **MR**) can be assigned to UUID f81d4fae-7de-c-11d0-a765-00a0c91e6bf6 by the vendor.

² Within this nomenclature, model-specific (also document-independent) features are referred to as "static" and document-specific (individual/personalized) features are referred to as "dynamic".

The following representation provides an overview of the generic basic check routine methodology. For this purpose, the three parts of the description of these check routines – light source, optical property of the characteristic, feature – are provided in the form of a table. The line, column and content of a cell represent the generic routine. The list might not be complete if check routines are missing, they can be added in accordance with the operator while following the format **(XX, YY, ZZ)** as specified above (the same applies for composite check routines).

Vendor- and model-specific check routines SHALL be classified and logged according to the schema given in ►Table 5.41

Properties or features of counterfeits SHOULD be incorporated by inverting the logic of check routines: e. g. a specific configuration of imitated security fibres by printing SHOULD be checked for absence of this pattern (i. e. VI, TR, IS).

Feature / Area (ZZ)		Light Source (XX)		
		VI	UV	IR
Fibers	FI		LU	
Full data page	FU		BR	
Static printed feature	IS	{AB, TR}	LU	{AB, TR, TL}
MRZ	MR	AB	BR	AB
Overprinted MRZ	OM		LU	
CAN	CA	AB		AB
Personalized perforation (dynamic)	PD	AB	LU	AB
Perforation of the substrate	PS	AB	LU	AB
Photo	PH	{AB, FR}	{BR, FR}	{AB, FR, TR}
Secondary Photo	SP	{AB, TR}	LU	{AB, TR}
Overprinted photo	OP		LU	
Security thread	TH	TR	LU	AB
Visual Inspection Zone (VIZ)	VZ		BR	
Watermark	WM			AB
Personalized dynamic feature	ID	{AB, TR}	LU	{AB, TR}
Additional feature	AF	{AB, BR, LU, TL, TR}	{AB, BR, LU, TL, TR}	{AB, BR, LU, TL, TR}

Table 5.41 Representation of the generic basic check routines

In addition, there can be the following composite check routines:

- **(IR, AB, TH)** in combination with **(VI, TR, TH)**: Check that a thread visible under IR light is invisible under white light.
- **(IR, TR, ID)** in combination with **(VI, AB, ID)**: Check that the ink of the dynamic print image absorbent under white light is transparent under IR light.
- **(IR, TR, IS)** in combination with **(IR, AB, IS)**: Check that some parts of the static print image are absorbent under IR light, whereas other parts of the characteristics are transparent.
- **(IR, TR, IS)** in combination with **(VI, AB, IS)**: Check that the ink of the static print image absorbent under white light is transparent under IR light.
- **(IR, TR, IS)** in combination with **(VI, AB, IS)** and **(IR, AB, ID)**: Check that the ink of the static print image absorbent under white light is transparent under IR light and a dynamically printed characteristic becomes visible in the same position under IR light.

5.7.2.3 Catalog for spectrally selective checks

Further information on the spectrally selective checks might be required by the operator for evaluation purposes and to update the underlying database to assure consistent and high quality authentication results over time. This information is the same for all documents of a specific document model, for example, textual explanations on the check routines, the image section from the reference database etc.

The vendor SHALL supply a catalog in machine readable form according to the defined Extensible Markup Language (XML) schema of this Technical Guideline which summarizes all necessary information on the spectrally selective verification checks. Due to the format, the catalog can be integrated into the evaluation of the results.

In addition, the catalog SHALL include the following information on each generic check routine of a document model:

- **UUID of the check routine:** This value specifies the unique id for the check.
- **Generic check identifier:** This attribute contains the generic check routine specification name, e. g. (**UV, BR, VZ**).
- **Proprietary check:** This attribute contains a vendor-specific name of the spectrally selective check routine (e. g. according to the entry in the check database).
- **Description:** This attribute contains a textual description of the check routine performed and, in particular, an explanation of the differences in the checks with multiple applications of one and the same generic specification.
- **Maximal range:** This attribute contains the upper limit of the range of values.
- **Minimal range:** This attribute contains the lower limit of the range of values.
- **Threshold:** This attribute contains the threshold configured for the decision for the spectrally selective verification.
- **Limits:**
 - **Bounding rectangle:** The bounding rectangle for the checking area if it is not a region.
 - **Region:** This element contains the description of an alternative geometric check region if a bounding rectangle is not applicable. The type description is freely definable by the vendor, the definition SHALL be disclosed to the operator.
- **Reference area:** This element contains the image section of the reference image region checked against the corresponding database.

For composite generic check routines, the function of which the decision is taken (so-called decision function) SHALL be specified. Any proprietary decision function SHALL be described by the vendor and disclosed to the operator. If it is not possible to describe the decision function technically, its functioning SHALL be described to be comprehensible.

The catalog SHALL list all test configurations. The test configurations assort the checks into check groups and include decision functions for the given document models. The checks for a specific model SHALL be conducted according to the configuration given in the catalog. For each configuration a unique identifier (UUID according to [RFC4122]) SHALL be assigned. Multiple configurations can exist for a given document model. The choice of a configuration is vendor specific.

All information contained in this catalog should be treated in a confidential manner and SHALL NOT be disclosed without the consent of the vendor.

5.7.3 Optical check result

The final optical result includes the results of all individual checks that were performed and SHALL be mapped according to ► Table 5.42.

Result of the check	Description
successful	The result of the check SHALL be set to successful if all checks prescribed in the subsections for the application scenario were successfully performed. The document does not show any optical/physical signs of falsification or any manipulations. The ICAO compliance check, see ▶Section 5.7.3.9, may result in not supported , but this SHALL not affect the final result.
failed	The result of the check SHALL be set to failed if at least one of the checks prescribed in the subsections for the application scenario was rated failed . The document deviates from the state and properties as supposed to be. It is suspected that the document has been manipulated.
undetermined	The optical checking process ended inconclusively.
not supported	The optical checking process could not be carried out for this document.
aborted	One of the individual checks was aborted due to an error.

Table 5.42 Final result of optical checks of an MRTD

5.7.3.1 Document detection and capturing

The correct placement of a document on the full page reader or a correct swipe through a swipe reader is crucial for all following steps.

For a full page reader, the inspection system SHALL provide mechanisms for automatic and manual triggering of document detection. Manual triggering is required if automatic document detection does not work properly. Image capturing SHOULD start automatically after the complete personal data page has been placed on the capturing surface. The firm- or software SHALL be able to compensate potential rotation and realign the image automatically and SHALL crop the captured data page accordingly for further processing. The presence of a document SHALL be detected only by using its optical properties. The detection process SHALL still be carried out optically, even if an expected chip is absent or malfunctioning.

In the case of a swipe reader, a document counts as detected when a MRZ could be read.

The result of the document model detection SHALL be mapped according to ▶Table 5.43.

Result	Description
successful	The document was placed correctly onto the reader and the process leads to a complete set of mandatory output images (full page reader) or MRZ characters (swipe reader).
failed	Detection failed, there might be no complete set of mandatory output images (full page reader) or MRZ characters (swipe reader) that can be used for document checks.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.43 Result of document detection

5.7.3.2 Document model identification

The correct identification of the document model is essential for all further checks. The IS SHALL implement mechanisms for automatic and manual triggering of the document model identification.

Manual selection SHALL be performed if the automatic document model identification fails and the system is operated by a qualified operator. Manual selection is not applicable in scenarios where the Inspection System is not operated by a qualified operator, e.g. self-service or automated border control scenarios.

The result of the document model detection SHALL be mapped according to ▶Table 5.44.

Result	Description
successful	The document model could be classified as specified in ▶Section 5.7.2.1.
failed	This value cannot occur here.

Result	Description
undetermined	Identifying the document model was not possible. If the operating conditions prohibit manual selection this means that the IS could not determine a document model. If manual selection is possible, this means that the IS could not determine a document model and that the operator did not choose any of the potential candidates for the document model that were presented to him by the IS.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.44 Result of the document model identification

The following value SHALL be logged:

- Document Model: The identified document model shall be provided in the format as specified in the format from ▶Section 5.7.2.1, i.e. (C, T, N, Y, P).

5.7.3.3 Check the document for outdated model

The IS SHALL check whether the identified document model was designated outdated by the inspection authority, for example, if it was declared invalid by the issuer. This check SHALL be performed under the condition that the document model identification was successful.

The result of this check SHALL be mapped according to ▶Table 5.45.

Result	Description
successful	The document model is not outdated.
failed	The document model is outdated.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.45 Result of checking for outdated document series

5.7.3.4 Check for test document

The Inspection Application SHALL check whether the document is not a test document.

The result of this check SHALL be mapped according to ▶Table 5.46.

Result	Description
successful	The document is not a test document.
failed	The document is a test document.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.46 Result of checking for test documents

5.7.3.5 Check if the document is only valid in the issuing country

The IS SHALL check if the document is only valid in the issuing country.

The result of this check SHALL be mapped according to ▶Table 5.47.

Result of the check	Description
successful	The chip is not only valid in the issuing country.
failed	The document is only valid in the issuing country.

Result of the check	Description
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.47 Result of checking if the document is only valid in the issuing country

5.7.3.6 Check for acknowledged issuer

The Inspection Application SHALL check whether the document has been issued by an issuer acknowledged by the inspection authority.

The result of this check SHALL be mapped according to ►Table 5.48.

Result	Description
successful	The document has been issued by an acknowledged issuer.
failed	The document has been issued by an issuer not acknowledged by the inspection authority.
undetermined	The inspection application could not determine the issuer of the document but an issuer was expected for this type of document.
not supported	The inspection application could not determine the issuer of the document and no issuer was expected for this type of document.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.48 Result of checking for acknowledged issuer

5.7.3.7 Check for the presence of the primary facial image

The Inspection Application SHALL check whether the primary facial image is present on the document.

The result of this check SHALL be mapped according to ►Table 5.49.

Result of the check	Description
successful	The primary facial image is present.
failed	The primary facial image is not present.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.49 Result of checking for the presence of the primary facial image

5.7.3.8 Check for the embedding of the primary facial image

The Inspection Application SHALL check whether the embedding of the primary facial image is as expected (e.g. glued, printed, ...).

The result of this check SHALL be mapped according to ►Table 5.50.

Result of the check	Description
successful	The embedding is as expected.
failed	The embedding is not as expected.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.50 Result of checking for the embedding of the primary facial image

5.7.3.9 Check the ICAO compliance of MRZ

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario. ICAO conformity checks SHALL use [ICAO9303].

The aggregation result SHALL be mapped according to ►Table 5.51.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.51 Final result of checking the MRZ for ICAO compliance

5.7.3.9.1 Check the consistency of MRZ data

An MRZ can be printed on a document. If it is present, it SHALL be read and the IS SHALL perform all checks given in ►Section 5.6.1: Check the MRZ data for the specific application scenario.

5.7.3.9.2 Check MRZ to be readable under IR light

The Inspection Application SHALL check whether the MRZ is present under an IR lighting source.

The result of this check SHALL be mapped according to ►Table 5.52.

Result	Description
successful	The MRZ is present under an IR lighting source.
failed	The MRZ is not present under an IR lighting source.
undetermined	This result SHALL NOT occur here.
not supported	No MRZ is expected for the identified document model.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.52 Result of checking the MRZ to be readable under IR light.

5.7.3.9.3 Check MRZ to be printed using OCR-B type face

The Inspection Application SHALL check whether the MRZ was printed using OCR-B type face according to [ISO/IEC 30116:2016].

The result of this check SHALL be mapped according to ►Table 5.53.

Result	Description
successful	The MRZ was printed using a OCR-B type face.
failed	The MRZ was not printed using a OCR-B type face.
undetermined	This result SHALL NOT occur here.
not supported	No MRZ is expected for the identified document model.

Result	Description
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.53 Result of checking the MRZ for OCR-B type face

5.7.3.9.4 Check alignment of MRZ

The Inspection Application SHALL check the alignment of the MRZ.

The result of this check SHALL be mapped according to ►Table 5.54.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.54 Result of checking the MRZ for alignment

Check character alignment of MRZ

The Inspection Application SHALL check whether the characters of MRZ were printed in alignment.

The result of this check SHALL be mapped according to ►Table 5.55.

Result	Description
successful	All characters of the MRZ were printed in alignment.
failed	The characters of the MRZ were not printed in alignment or an ICAO-compliant MRZ is expected to be present for the identified document model but is not present.
undetermined	This value cannot occur here.
not supported	This value cannot occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.55 Result of checking the MRZ for character alignment

Check horizontal alignment of MRZ

The Inspection Application SHALL check the horizontal alignment of the MRZ.

The result of this check SHALL be mapped according to ►Table 5.56.

Result	Description
successful	The MRZ was printed in horizontal alignment.
failed	The MRZ was not printed in horizontal alignment or an ICAO-compliant MRZ is expected to be present for the identified document model but is not present.
undetermined	This value cannot occur here.
not supported	This value cannot occur here.

Result	Description
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.56 Result of checking the MRZ for horizontal alignment

Check vertical alignment of MRZ

The Inspection Application SHALL check the vertical alignment of the MRZ.

The result of this check SHALL be mapped according to ►Table 5.57.

Result	Description
successful	The MRZ was printed in vertical alignment.
failed	The MRZ was not printed in vertical alignment or an ICAO-compliant MRZ is expected to be present for the identified document model but is not present.
undetermined	This value cannot occur here.
not supported	This value cannot occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.57 Result of checking the MRZ for vertical alignment

5.7.3.10 VIZ consistency check

The IS SHALL check the VIZ for consistency. This means that the IS shall check whether the information in the VIZ is represented as specified in [ICAO9303] Eighth Edition Parts 3-5.

The result of this check SHALL be mapped according to ►Table 5.58.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.58 Result of checking VIZ for consistency

5.7.3.10.1 VIZ's issuer versus VIZ's nationality consistency check

The IS SHALL check whether the issuer and the nationality are consistent with each other.

Notes:

1. The logic of this check depends on the document type (e.g. visa vs. passport)
2. Even for passports it is not sufficient to check for equality because certain countries may issue passports for non-nationals or may differentiate between citizenship and nationality.

The result of this check SHALL be mapped according to ►Table 5.59.

Result	Description
successful	The issuer and nationality are consistent with each other.

Result	Description
failed	The issuer and nationality are not consistent with each other.
undetermined	This result SHALL NOT occur here.
not supported	The VIZ does not contain the issuer or/and the nationality.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.59 Result of checking VIZ's consistency between issuer and nationality

5.7.3.10.2 VIZ's document type consistency check

The IS SHALL check whether the document type given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ► Table 5.60.

Result	Description
successful	The document type is ICAO compliant.
failed	The document type is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.60 Result of checking VIZ's document type

5.7.3.10.3 VIZ's issuer consistency check

The IS SHALL check whether the issuer given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

Whether the issuer is included in the ICAO list of known nationalities (i.e. country codes) SHALL NOT be part of this check.

The result of this check SHALL be mapped according to ► Table 5.61.

Result	Description
successful	The issuer is ICAO compliant.
failed	The issuer is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.61 Result of checking VIZ's issuer

5.7.3.10.4 VIZ's nationality consistency check

The IS SHALL check whether the nationality given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

Whether the nationality is included in the ICAO list of known nationalities (i.e. country codes) SHALL NOT be part of this check.

The result of this check SHALL be mapped according to ► Table 5.62.

Result	Description
successful	The nationality is ICAO compliant.
failed	The nationality is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.

Result	Description
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.62 Result of checking VIZ's nationality

5.7.3.10.5 VIZ's date of birth consistency check

The IS SHALL check whether the date of birth given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.63.

Result	Description
successful	The date of birth is ICAO compliant.
failed	The date of birth is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.63 Result of checking VIZ's date of birth

5.7.3.10.6 VIZ's date of expiry consistency check

The IS SHALL check whether the date of expiry given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.64.

Result	Description
successful	The date of expiry is ICAO compliant.
failed	The date of expiry is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.64 Result of checking VIZ's date of expiry

5.7.3.10.7 VIZ's sex consistency check

The IS SHALL check whether the sex given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.65.

Result	Description
successful	The sex is ICAO compliant.
failed	The sex is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.65 Result of checking VIZ's sex

5.7.3.10.8 VIZ's document number consistency check

The IS SHALL check whether the document number given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.66.

Result	Description
successful	The document number is ICAO compliant.
failed	The document number is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.66 Result of checking VIZ's document number

5.7.3.10.9 VIZ's surname consistency check

The IS SHALL check whether the surname given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ► Table 5.67.

Result	Description
successful	The surname is ICAO compliant.
failed	The surname is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.67 Result of checking VIZ's surname

5.7.3.10.10 VIZ's given name consistency check

The IS SHALL check whether the given name given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ► Table 5.68.

Result	Description
successful	The given name is ICAO compliant.
failed	The given name is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.68 Result of checking VIZ's given name

5.7.3.10.11 VIZ's issuer ICAO included check

The IS SHALL check whether the issuer (country code) is included in the list of codes in [ICAO9303] Eighth Edition Part 3.

The result of this check SHALL be mapped according to ► Table 5.69.

Result	Description
successful	The issuer is an ICAO included country code.
failed	The issuer is not an ICAO included country code.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.69 Result of checking VIZ's issuer (ICAO included)

5.7.3.10.12 VIZ's nationality ICAO included check

The IS SHALL check whether the nationality (country code) is included in the list of codes in [ICAO9303] Eighth Edition Part 3.

The result of this check SHALL be mapped according to ►Table 5.70.

Result	Description
successful	The nationality is an ICAO included country code.
failed	The nationality is not an ICAO included country code.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.70 Result of checking VIZ's nationality (ICAO included)

5.7.3.10.13 VIZ's authority or issuing organization consistency check

The IS SHALL check whether the authority or issuing organization given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.71.

Result	Description
successful	The authority or issuing organization is ICAO compliant.
failed	The authority or issuing organization is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.71 Result of checking VIZ's given name

5.7.3.10.14 VIZ's Card Access Number (CAN) consistency check

The IS SHALL check whether the CAN given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.72.

Result	Description
successful	The card access number is ICAO compliant.
failed	The card access number is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.72 Result of checking VIZ's CAN

5.7.3.10.15 VIZ's date of issue consistency check

The IS SHALL check whether the date of issue given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.73.

Result	Description
successful	The date of issue is ICAO compliant.
failed	The date of issue is not ICAO compliant.

Result	Description
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.73 Result of checking VIZ's date of issue

5.7.3.10.16 VIZ's place of issue consistency check

The IS SHALL check whether place of issue given in the visual zone is ICAO compliant. ICAO compliance SHALL be interpreted as complying with [ICAO9303].

The result of this check SHALL be mapped according to ►Table 5.74.

Result	Description
successful	The place of issue is ICAO compliant.
failed	The place of issue is not ICAO compliant.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.74 Result of checking VIZ's place of issue

5.7.3.11 Check for the consistency of MRZ and VIZ

The IS SHALL check whether the information in the MRZ and VIZ are consistent with each other by executing a component check as described in ►Section 5.6.4 for each individual element (first name, last name, date of birth, sex, date of expiry, optional data, etc.) expected to be included in both sources for the identified document model.

Since the VIZ may contain non-Latin characters, it could be necessary to perform a transliteration suitable for the respective document for the comparison. The rules of transliteration and truncation SHOULD comply with [ICAO9303].

In the first attempt, the VIZ SHALL be analysed by OCR from the IR record. If the attempt fails to extract the data correctly, a white light (VI) image SHALL be used as basis for OCR.

The result of this check SHALL be mapped according to ►Table 5.75.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.75 Result of checking for the consistency of MRZ and VIZ

5.7.3.12 Spectrally selective verification

Most MRTDs can be checked regarding their authenticity by verifying their spectrally selective security features. The specific spectrally selective check routines being performed depend on the document model. One exception is (IR, AB, MR) (e. a. B900), which can be parametrized independently from a specific document model. With certain restrictions on accuracy, (UV, BR, FU) can also be performed without selecting a specific document model.

Please refer to [BSI TR-03135-2] for mandatory checks.

5.7.3.12.1 Optical document model database

Because checks are document dependent, using a dedicated database tailored to the specific check is RECOMMENDED in all cases. This database SHALL be updated regularly in order to reduce false acceptance and rejection rates. It is RECOMMENDED to specify and maintain the contents of the database together with the operator; in most cases, this includes:

- Definition of model identifiers in order to differentiate the document models.
- Definition of search areas and their contents as well as the checks based on them including thresholds and ranges of values.
- Subsequent extension, correction and readjustment of the specified checks in order to be able to take into account any experience gained with respect to the variances in the features of authentic documents to be checked (e. g. regarding aging effects, traces of usage, variability in the production process).

The specified checks, as well as their associated thresholds, proprietary decision functions, etc. SHALL be documented and disclosed by the vendor.

5.7.3.12.2 Hierarchical organization of checks

There are three levels in the hierarchy of spectrally selective checks:

- Basic and composite checks,
- group checks, and
- the final spectrally selective verification result.

This means that individual spectrally selective checks (including composite checks) are organized into groups which contain the assigned basic checks. Groups of such checks are then aggregated into the final spectrally selective verification result.

5.7.3.12.3 Perform Spectrally Selective Verification

The IS SHALL perform spectrally selective verification on the document.

The final check result for spectrally selective checks SHALL aggregate all spectrally selective check group results of all checks prescribed in the subsections for the application scenario.

The final result of this check SHALL be mapped according to ►Table 5.76.

Result of the check	Description
successful	A decision function yields a positive result incorporating all spectrally selective check groups on a specific document.
failed	A decision function yields a negative result incorporating all spectrally selective check groups on a specific document.
undetermined	A decision function yields an undetermined result incorporating all spectrally selective check groups on a specific document.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow. For example, one of the individual checks was aborted due to an error.

Table 5.76 Final result of performing Spectrally Selective Verification

Check spectrally selective verification group check

The IS SHALL aggregate the results of basic checks (see ▶Section 5.7.3.12.3.1: Check basic spectrally selective checks) and composite checks (see ▶Section 5.7.3.12.3.1: Check composite spectrally selective checks) in group checks.

For this, all spectrally selective (basic or composite) checks SHALL be grouped even if a group only consists of one check.

The result of this check SHALL be mapped according to ▶Table 5.77.

Result of the check	Description
successful	A decision function yields a positive result incorporating all spectrally selective basic (or composite) checks in this particular group.
failed	A decision function yields a negative result incorporating all spectrally selective basic (or composite) checks in this particular group.
undetermined	A decision function yields an undetermined result incorporating all spectrally selective basic (or composite) checks in this particular group.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.77 Result of a spectrally selective verification group check

For each spectrally selective check group, the following values SHALL be logged:

- UUID of the group: This value specifies the unique id for the check group.
- Group name: A textual description of what this check group represents / does. The description should be clear enough to be used by an IS to show it to an end-user.
- Involved Checks: A list of UUIDs of the spectrally selective checks referenced in this group.

Check basic spectrally selective checks

The IS SHALL perform a **particular** spectrally selective check routine for a specific document model.

The result of this check SHALL be mapped according to ▶Table 5.78.

Result of the check	Description
successful	A particular spectrally selective security feature to be checked for the given document was successfully tested by a decision function with a positive result.
failed	A particular spectrally selective security feature to be checked for the given document was successfully tested by a decision function with a negative result.
undetermined	A particular spectrally selective security feature to be checked for the given document was successfully tested by a decision function with an undetermined result.
not supported	This result SHALL not occur here. An individual check will only be performed on a document that is expected to support the check.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.78 Result of performing basic spectrally selective checks

For each basic check routine, the following values SHALL be logged:

- UUID of the check routine: This value specifies the unique id for the check.
- Generic check identifier: This attribute contains the generic check routine specification name, e. g. (UV, BR, VZ).
- Proprietary check: This attribute contains a vendor-specific name of the spectrally selective check routine (e. g. according to the entry in the check database).
- Result: This attribute represents the result of the current spectrally selective check. The assignment of this attribute is performed based on ▶Table 5.78.
- Margin: This attribute contains a standardized result value between -1 and +1, referring to a standardized threshold of 0. Here, a value greater or equal 0 refers to the result "successful" and a value less than 0 to the result "failed". The vendor is responsible for the detailed calculation of the margin.

- Scanned area:
 - Bounding rectangle: The bounding rectangle for the checking area if it is not a region.
 - Region: This element contains the description of an alternative geometric check region if a bounding rectangle is not applicable. The type description is freely definable by the vendor, the definition SHALL be disclosed to the operator.
 - Image area: This element contains the image section from the live record of the checked document area.

Check composite spectrally selective checks

The IS SHALL perform spectrally selective check routines that combine basic checks into composite checks for a specific document model. Hence, checks of different spatial and/or spectral regions can be combined to more complex, i.e. composite checks.

The result of this aggregated check SHALL be mapped according to

►Table 5.79.

Result of the check	Description
successful	A composite spectrally selective security feature to be checked for the given document was successfully tested by a decision function with a positive result.
failed	A composite spectrally selective security feature to be checked for the given document was successfully tested by a decision function with a negative result.
undetermined	A composite spectrally selective security feature to be checked for the given document was successfully tested by a decision function with an undetermined result.
not supported	This result SHALL not occur here. A composite check will only be performed on a document for which the underlying basic checks are performed.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.79 Result of performing composite spectrally selective checks

The log entry SHALL refer to the UUIDs of the used basic check routines.

5.8 Checking an electronic component

The following sections describe how an IS SHALL inspect an electronic component.

The specific checks to be performed depend on the application scenario and SHALL be defined in a profiling document.

5.8.1 Aggregation of electronic check results

The electronic check consists of several partial checks that are aggregated together. The previously specified aggregation rules from ►Section 3.8 are applied to all partial check results. The associated return value is translated into traffic light colours according to ►Table 5.1 and displayed as the result of the electronic check. The aggregation of these check results is shown in the table ►Table 5.80 below. Each row describes an electronic check, which is divided into partial checks. A further subdivision of the check is described after each further column to the right.

Document Check Type	Partial Checks	Partial Checks	Partial Checks	Partial Checks
Electronic Check Checks	►Section 5.8.2: Check for the presence of applications SelectableApplicationsCheck	►Section 5.8.2.1: Check the presence of one specific application SelectableApplicationCheck		

Document Check Type	Partial Checks	Partial Checks	Partial Checks	Partial Checks
	▶Section 5.8.3: Check for chip access AccessProtocol-sCheck	▶Section 5.8.3.1: Check for chip access performing BAC BacCheck		
		▶Section 5.8.3.2: Check for chip access performing PACE PaceCheck		
		▶Section 5.8.3.3: Check for chip access without a secure protocol PlainCheck		
	▶Section 5.8.4: Check the mandatory files MandatoryFile-sCheck	▶Section 5.8.4.1: Check the availability of mandatory files AvailabilityCheck	▶Section 5.8.4.1.1: Check the availability of a mandatory file FileAvailabilityCheck	
		▶Section 5.8.4.2: Check the syntax of mandatory files SyntaxCheck	▶Section 5.8.4.2.1: Check EF.SOD EfSodCheck	▶Section 5.8.4.2.1.1: Check the data group hash list DataGroupHashList-Check
				▶Section 5.8.4.2.1.2: Check the signature section SignatureSectionCheck
				▶Section 5.8.4.2.1.3: Check the document signer certificate DocumentSignerCertificateCheck
			▶Section 5.8.4.2.2: Check EF.COM EfComCheck	▶Section 5.8.4.2.2.1: Check the LDS version LdsVersionCheck
				▶Section 5.8.4.2.2.2: Check the Unicode version UnicodeVersionCheck
				▶Section 5.8.4.2.2.3: Check the data group list DataGroupListCheck
		▶Section 5.8.4.2.3: Check EF.DG1 Dg1Check		Partial checks see ▶Table 5.4.

Document Check Type	Partial Checks	Partial Checks	Partial Checks	Partial Checks
			▶Section 5.8.4.2.4: Check EF.DG2 Dg2Check	▶Section 5.8.4.2.4.1: Check the biometric header (EF.DG2) BiometricHeader-Check ▶Section 5.8.4.2.4.2: Check the image (EF.DG2) ImageCheck
	▶Section 5.8.5: Check the authenticity AuthenticityCheck	▶Section 5.8.5.1: Active Authentication Check ActiveAuthCheck		
		▶Section 5.8.5.2: Chip Authentication Check ChipAuthCheck		
		▶Section 5.8.5.3: PACE Chip Authentication Mapping Check PaceChipAuthMapping-Check		
	▶Section 5.8.6: Check the signature SignatureCheck	▶Section 5.8.6.1: Check an individual security file SecurityFileCheck	▶Section 5.8.6.1.1: Check the digest of the security file DigestCheck	
			▶Section 5.8.6.1.2: Check the signature value of a security file SignatureValueCheck	
	▶Section 5.8.7: Check the issuer certificates IssuerCertificate-Check	▶Section 5.8.7.1: Check a certificate of a security file SecurityFileCheck	Partial checks see ▶Table 5.29.	
	▶Section 5.8.8: Check the integrity of the electronic component IntegrityCheck	▶Section 5.8.8.1: Check the consistency between EF.SOD and EF.COM EfSodVsEfComCheck		
		▶Section 5.8.8.2: Check the integrity of the data groups DataGroupsIntegrity-Check	▶Section 5.8.8.2.1: Check the integrity of a single data group DataGroupIntegrity-Check	
	▶Section 5.8.9: Check the issuing country IssuingCountryCheck	▶Section 5.8.9.1: Check a signer certificate SignerCertificate-Check	Partial checks see ▶Table 5.24.	

Table 5.80 Aggregation of electronic document check results

5.8.2 Check for the presence of applications

While inspecting the electronic component, the IS requires the presence of the corresponding application(s), respectively Dedicated Files (DFs). Which applications need to be present is dependent on the specific document model. In any case, at least the ICAO eMRTD application SHALL be present.

The IS SHALL aggregate the results of the check prescribed in the following subsection, which is executed for each expected application, for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.81.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.81 Result of checking for the presence of applications

5.8.2.1 Check the presence of one specific application

The IS SHALL check whether a specific application is present or missing. For this, the application SHALL be selectable by the reader.

The aggregation result SHALL be mapped according to ►Table 5.82.

Result	Description
successful	The application is present.
failed	The application is missing.
undetermined	This result SHALL not occur here.
not supported	This result SHALL not occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.82 Result of checking the presence of one specific application

5.8.3 Check for chip access

Performing access protocols is in the majority of cases a necessary step for a complete electronic document check. For certain document models, a failure or unavailability of a specific access protocol can be an indicator for a manipulated or counterfeit eMRTD.

As described in ►Section 5.8.10, either Basic Access Control (BAC) or Password Authenticated Connection Establishment (PACE) SHALL be performed in order to ensure secure and confidential chip access via the contactless communication channel, unless the document supports neither BAC nor PACE.

The individual results of the checks can be found in the following subsections and are aggregated into the following final result. It is possible that a document supports both PACE and BAC. In this case the Inspection Application SHOULD offer a configuration option which protocol to use in preference.

The final result SHALL be mapped according to ►Table 5.83.

Result of the check	Description
successful	Either BAC or PACE could be performed successfully or the chip could be accessed without a secure protocol.
failed	None of the secure protocols (BAC or PACE) were performed successfully and the chip could also not be accessed without a secure protocol.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	One of the individual checks was aborted due to an error.

Table 5.83 Final result of checking the chip access

5.8.3.1 Check for chip access performing BAC

The result of accessing the chip using BAC SHALL be mapped according to ►Table 5.84.

Result of the check	Description
successful	BAC could be performed successfully.
failed	BAC was performed, but failed. The secure messaging channel could not be established successfully.
undetermined	This result SHALL NOT occur here.
not supported	The document does not support BAC.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.84 Result of performing the access protocol BAC

5.8.3.2 Check for chip access performing PACE

The result of accessing the chip using PACE SHALL be mapped according to ►Table 5.85.

Furthermore, the used version, mode (CAM, GM or IM), algorithms (asymmetric and symmetric) and domain parameters SHALL be logged.

Result of the check	Description
successful	PACE could be performed successfully.
failed	PACE was performed, but failed. The secure messaging channel could not be established successfully.
undetermined	This result SHALL NOT occur here.
not supported	The document does not support PACE.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.85 Result of performing the access protocol PACE

5.8.3.3 Check for chip access without a secure protocol

The result of accessing the chip without a secure protocol SHALL be mapped according to ►Table 5.86. In general, this SHALL be the last option the IS tries, after the chip access with BAC and, if applicable, PACE has failed.

Result of the check	Description
successful	The chip could be accessed without a secure protocol.
failed	The chip could not be accessed without a secure protocol.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.86 Result of accessing the chip without a secure protocol

5.8.4 Check the mandatory files

Mandatory files are those that are required to be present for the document under inspection according to the relevant standard. For eMRTDs, the standard is usually [ICAO9303], where detailed information on mandatory files can be found in [ICAO9303] Part 10.

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.87

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.87 Result of checking the mandatory files

5.8.4.1 Check the availability of mandatory files

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.88.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.88 Result of checking the mandatory files

5.8.4.1.1 Check the availability of a mandatory file

The IS SHALL check whether a specific mandatory file is present or missing. For this, the file SHALL be selectable by the reader.

The result of this check SHALL be mapped according to ►Table 5.89.

Result of the check	Description
successful	The file is present.

Result of the check	Description
failed	The file is missing.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.89 Result of checking the availability of a mandatory file

5.8.4.2 Check the syntax of mandatory files

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.90.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.90 Result of checking the mandatory files

5.8.4.2.1 Check EF.SOD

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.91.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.91 Result of checking the syntax of EF.SOD

Check the data group hash list

The IS SHALL check whether the data group hash list is syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.92.

Result of the check	Description
successful	The data group hash is syntactically correct.
failed	The data group hash is syntactically incorrect (e.g. missing entries for present data groups).
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.92 Result of checking the data group hash list

Check the signature section

The IS SHALL check whether the signature section is present and syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.93.

Result of the check	Description
successful	A signature section is present and syntactically correct.
failed	A signature section is not present or syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.93 Result of checking the signature section

Check the document signer certificate

The IS SHALL check whether a document signer certificate that matches the signer identifier of the signature is present and syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.94.

Result of the check	Description
successful	A document signer certificate matching the signer identifier is present and syntactically correct.
failed	A document signer certificate matching the signer identifier is not present or syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.94 Result of checking the document signer certificate

5.8.4.2.2 Check EF.COM

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.95.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.

Result	Description
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.95 Result of checking the syntax of EF.COM

Check the LDS version

The IS SHALL check whether the LDS version is present in the EF.COM and syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.96.

Result of the check	Description
successful	The LDS version is present and syntactically correct.
failed	The LDS version is not present or syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.96 Result of checking the LDS version

Check the Unicode version

The IS SHALL check whether the Unicode version is present and syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.97.

Result of the check	Description
successful	The Unicode version is present and syntactically correct.
failed	The Unicode version is not present or syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.97 Result of checking the Unicode version

Check the data group list

The IS SHALL check whether the data group list is present and syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.98.

Result of the check	Description
successful	The data group list is present and syntactically correct.
failed	The data group list is not present or syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.98 Result of checking the data group list

5.8.4.2.3 Check EF.DG1

EF.DG1 contains MRZ data.

The IS SHALL perform all checks given in ►Section 5.6.1: Check the MRZ data for the specific application scenario.

5.8.4.2.4 Check EF.DG2

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.99.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.99 Result of checking the syntax of EF.DG1

Check the biometric header (EF.DG2)

The IS SHALL check whether the first biometric information template has a header that is syntactically correct.

The result of this check SHALL be mapped according to ►Table 5.100.

Result of the check	Description
successful	The first biometric information template is present and has a header that is syntactically correct.
failed	The first biometric information template is not present and has no header that is syntactically incorrect.
undetermined	This result SHALL NOT occur here.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.100 Result of checking the biometric header (EF.DG2)

Check the image (EF.DG2)

The IS SHALL check whether the first biometric information template contains an image that is syntactically correct, is in a format that is allowed by the ICAO and is not empty.

The result of this check SHALL be mapped according to ►Table 5.101.

Result of the check	Description
successful	The first biometric information template contains an image that is syntactically correct, is in a format that is ICAO compliant and is not empty.
failed	The first biometric information template contains an image that is syntactically incorrect, is not in a format that is ICAO compliant or is empty.
undetermined	This result SHALL NOT occur here.

Result of the check	Description
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.101 Result of checking the image (EF.DG2)

5.8.5 Check the authenticity

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

In order to check the authenticity of the electronic component, the protocols Active Authentication (AA), Chip Authentication (CA), or Password Authenticated Connection Establishment - Chip Authentication Mapping (PACE-CAM) can be performed.

The aggregation result SHALL be mapped according to

►Table 5.102.

Result	Description
successful	Either AA, CA, or PACE-CAM are successful.
failed	All three protocols were performed and failed, or all three protocols were performed and one failed and the others were NOT successful.
undetermined	The process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, successful or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.102 Result of checking the authenticity

5.8.5.1 Active Authentication Check

The IS SHALL check whether the AA protocol (specified in [ICAO9303]) is supported and how it was performed.

The result of this check SHALL be mapped according to ►Table 5.103

Result of the check	Description
successful	The protocol could be performed successfully.
failed	The protocol was performed, but failed, although the required data and parameters are available.
undetermined	The data contained in the document suggests that it supports the protocol, but it could not be executed because of missing data or parameters.
not supported	The document does not support the protocol.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.103 Result of checking the authenticity via Active Authentication

If the defect Active Authentication Private Keys Compromised applies to the checked document, the requirements specified in ►Section 5.12.2.1.5 SHALL be implemented.

If the defect Authentication Protocol Failure with status code AA_failure or CA_AA_failure (►Section 5.12.2.1.6) is detected, the check SHALL be performed, and if the returned check result is failed, it SHALL be set to undetermined.

5.8.5.2 Chip Authentication Check

The IS SHALL check whether the CA protocol (specified in [BSI TR-03110]) is supported and how it was performed.

The result of this check SHALL be mapped according to ►Table 5.104

Result of the check	Description
successful	The protocol could be performed successfully.
failed	The protocol was performed, but failed, although the required data and parameters are available.
undetermined	The data contained in the document suggests that it supports the protocol, but it could not be executed because of missing data or parameters.
not supported	The document does not support the protocol.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.104 Result of checking the authenticity via Active Authentication

If the defect `Chip Authentication Private Keys Compromised` applies to the checked document, the requirements specified in section ►Section 5.12.2.1.4 SHALL be implemented.

If the defect `Authentication Protocol Failure` with status code `CA_failure` or `CA_AA_failure` (►Section 5.12.2.1.6) is detected, the check SHALL be performed, and if the returned check result is `failed`, it SHALL be set to `undetermined`.

5.8.5.3 PACE Chip Authentication Mapping Check

The IS SHALL check whether the chip authenticity part of the PACE-CAM protocol (as specified in [ICAO9303]) is supported and how it was performed. For this check it is only relevant whether the authenticity of the electronic component could be verified by using PACE-CAM.

The result of this check SHALL be mapped according to ►Table 5.105

Result of the check	Description
successful	The protocol could be performed successfully.
failed	The protocol was performed, but failed, although the required data and parameters are available.
undetermined	The data contained in the document suggests that it supports the protocol, but it could not be executed because of missing data or parameters.
not supported	The document does not support the protocol.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.105 Result of checking the authenticity via PACE

If the defect `Chip Authentication Private Keys Compromised` applies to the checked document, the requirements specified in ►Section 5.12.2.1.4 SHALL be implemented.

If the defect `Authentication Protocol Failure` with status code `PACE_CAM_failure` (see ►Section 5.12.2.1.6) is detected, the check SHALL be performed and, if the returned check result is `failed`, it SHALL be set to `undetermined`.

5.8.6 Check the signature

For each security object:

- `EF.SOD` (mandatory)
- `EF.CardSecurity` (conditional: mandatory if read)
- `EF.ChipSecurity` (conditional: mandatory if read)

the hash and signature SHALL be checked according to ►Section 5.8.6.1).

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.106.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.106 Result of checking the signatures

5.8.6.1 Check an individual security file

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.107.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.107 Result of checking an individual security file

If the defect Document Security Object Malformed applies to the checked document, the requirements specified in ►Section 5.12.2.2.2 SHALL be implemented.

If the defect Card Security Object Malformed applies to the checked document, the requirements specified in ►Section 5.12.2.3.1 SHALL be implemented.

If the defect Chip Security Object Malformed applies to the checked document, the requirements specified in ►Section 5.12.2.3.2 SHALL be implemented.

5.8.6.1.1 Check the digest of the security file

The IS SHALL compare the digest contained inside the security file with the digest calculated by the IS.

The result of this check SHALL be mapped according to ►Table 5.108

Result of the check	Description
successful	The two comparands are equal.

Result of the check	Description
failed	The two comparands are not equal.
undetermined	The comparison could not be executed. For example, because the hash algorithm used is not known to the IS.
not supported	There is no digest inside the security file.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.108 Result of checking the digest of the security file

5.8.6.1.2 Check the signature value of a security file

The IS SHALL check the signature.

The result of this check SHALL be mapped according to

►Table 5.109

Result of the check	Description
successful	The signature verification of the specific security object was performed successfully.
failed	The signature verification of the specific security object failed.
undetermined	Information is missing to perform the signature verification (e. g. the signer certificate).
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.109 Result of checking the signature of the security file

5.8.7 Check the issuer certificates

For each existing security object

- EF.SOD (mandatory)
- EF.CardSecurity (conditional: mandatory if read)
- EF.ChipSecurity (conditional: mandatory if read)

the issuer certificates SHALL be checked according to ►Section 5.9.1.4. For this purpose, the requirement for the architecture of the Inspection System defined in chapter ►Section 4.2 SHALL be implemented.

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.110.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.110 Result of checking the issuer certificates

5.8.7.1 Check a certificate of a security file

The IS SHALL perform the checks given in ▶Section 5.6.3.2 for the certificate contained inside the security file.

5.8.8 Check the integrity of the electronic component

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to

▶Table 5.111

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.111 Result of checking the mandatory files

5.8.8.1 Check the consistency between EF.SOD and EF.COM

The IS SHALL check the consistency between EF.SOD and EF.COM. This check is necessary because the EF.COM is not secured by a digital signature.

The result of this check SHALL be mapped according to ▶Table 5.112

Result of the check	Description
successful	The contents of EF.COM and EF.SOD are consistent with each other.
failed	The contents of EF.COM and EF.SOD are not consistent with each other.
undetermined	This result SHALL NOT occur here.
not supported	One or both files are not present.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.112 Result of checking the consistency of EF.SOD and EF.COM

If the defect EF.COM and EF.SOD discrepancy applies to the checked document, the requirements specified in section ▶Section 5.12.2.2.3 SHALL be implemented.

5.8.8.2 Check the integrity of the data groups

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according ▶Table 5.113

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.

Result	Description
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.113 Result of checking the integrity of the data groups

If the defect Data Group Malformed is included in the Defect List and can be identified in a specific data group, the requirements specified in ▶Section 5.12.2.2.1 SHALL be implemented.

5.8.8.2.1 Check the integrity of a single data group

The IS SHALL check the integrity of the data groups. The hashes in the EF . SOD are the basis for this check. This check SHALL be executed for every data group that is read.

The result of this check SHALL be mapped according to ▶Table 5.114

Result of the check	Description
successful	The calculated hash and the one stored in EF . SOD are identical.
failed	The calculated hash and the one stored in EF . SOD are not identical.
undetermined	The hash cannot be calculated because the hash algorithm is not supported by the inspection system.
not supported	This result SHALL NOT occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.114 Result of checking the integrity of a single data group

5.8.9 Check the issuing country

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ▶Table 5.115.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.115 Result of checking the issuing countries

5.8.9.1 Check a signer certificate

The IS SHALL perform the checks given in ►Section 5.6.2.2 for each certificate contained in a security file.

5.8.10 Access and process sequences

Accessing and inspecting the electronic data is accomplished by executing the access and security protocols defined in [ICAO9303] and [BSI TR-03110]. It is RECOMMENDED that Inspection Systems support the following protocols:

- Basic Access Control (BAC) according to [ICAO9303]
- Password Authentication Connection Establishment (PACE) according to [ICAO9303]
- Passive Authentication (PA) according to [ICAO9303]
- Active Authentication (AA) according to [ICAO9303]
- Chip Authentication Version 1 (CAv1) according to [BSI TR-03110]
- Chip Authentication Version 2 (CAv2) according to [BSI TR-03110]
- Terminal Authentication Version 1 (TAv1) according to [BSI TR-03110]
- Terminal Authentication Version 2 (TAv2) according to [BSI TR-03110]

The profiling SHOULD specify which protocols and security mechanisms are mandatory depending on the application scenario.

In order to perform these protocols, an Inspection System SHOULD be able to read and process at least the following files, if available on the document according to the specifications, and if access is granted:

- **Deprecated:** EF.COM (index indicating which data groups are stored in the chip)
- EF.SOD (hash values of all data groups as well as the electronic signature over these hash values)
- EF.DG14 (security information, especially Chip Authentication Public Key)
- EF.DG15 (Active Authentication Public Key)
- EF.CVCA (Country Verifying CA reference(s))
- EF.CardAccess (security information according to [ICAO9303])
- EF.CardSecurity (security information according to [ICAO9303])
- EF.ChipSecurity (security information according to [BSI TR-03110])
- EF.DG1 (contains the biographic data of the document owner)
- EF.DG2 (contains the biometric facial image)
- EF.DG3 (contains the fingerprints)

The profiling SHOULD specify which files to read and process.

In the following sections it is assumed that the information (MRZ, CAN) required for establishing a secure communication channel (BAC, PACE) is available, in order to perform the protocols.

The following diagrams and sections show and describe how an IS can set up the process of accessing and reading the chip using different protocols and their options. An overview of the process is shown in ►Figure 5.3. Detailed diagrams and descriptions of the subprocesses are given in the subsequent sections.

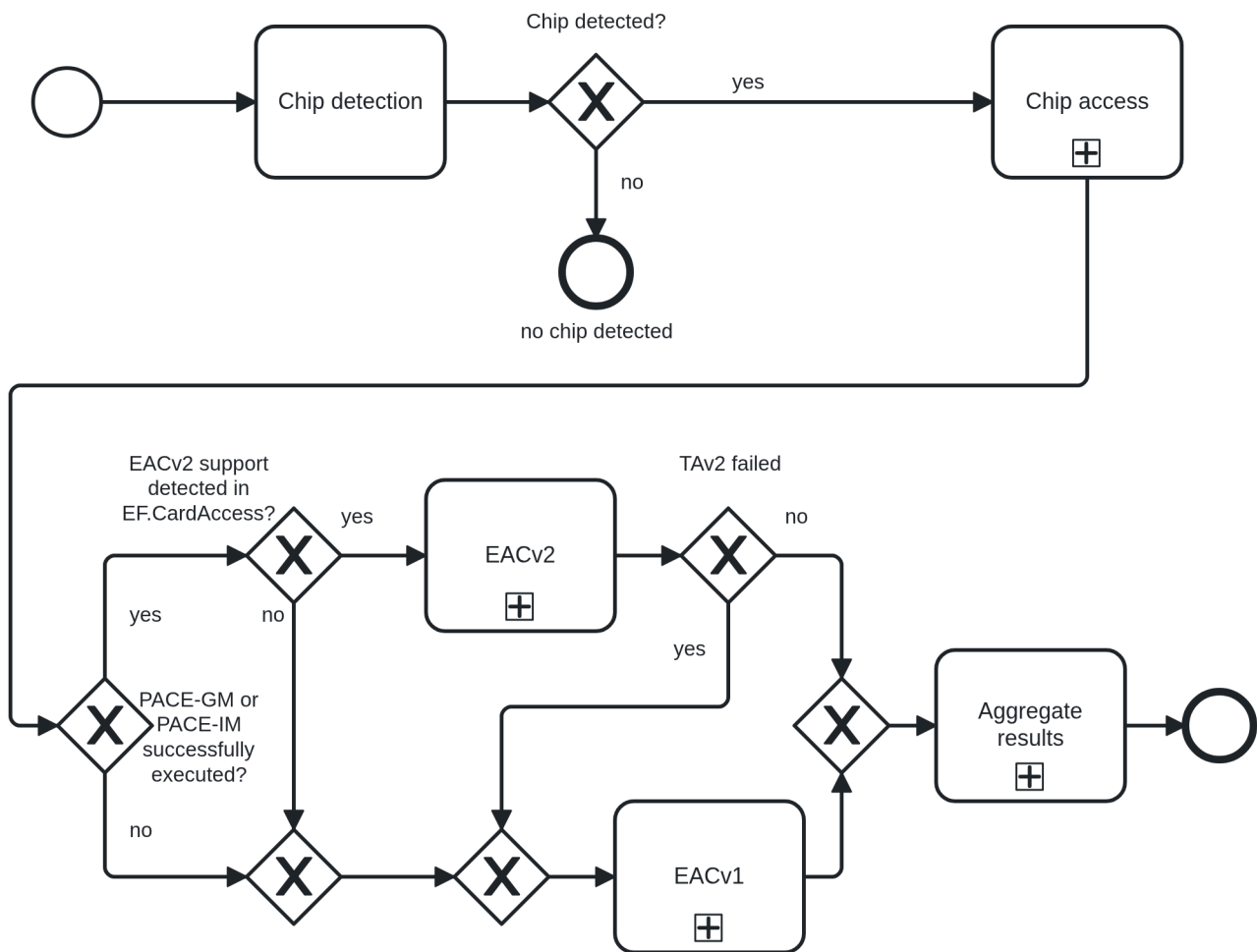


Figure 5.3. Overview of electronic chip access process

The IS SHALL establish whether or not an MRTD possesses an electronic chip. Provided that a chip has been detected, the IS SHALL attempt to establish a secure communication channel with the chip using chip access protocols (BAC or PACE), see ▶Section 5.8.10.1.

The chip reading process is completed by executing EAC in order to ensure the authenticity of the chip and of the terminal, and to read the information stored in the data groups of the chip. The IS SHALL automatically choose the appropriate version of the EAC depending on the result of the chip access subprocess as described in ▶Section 5.8.10.1.

EAC consists of several subprocesses, including terminal authentication, chip authentication, and data group access, which are carried out sequentially. Those constituent subprocesses of EAC are described in ▶Section 5.8.10.2.

After the EAC subprocess has been completed, the IS SHALL aggregate and log the results of the performed protocols and checks, see ▶Section 5.8.10.3.

5.8.10.1 Chip access protocols

This section covers the "Chip Access" subprocess of ▶Figure 5.3. A detailed view of this subprocess is shown in ▶Figure 5.4. In this case, the access protocols are either BAC or a PACE variant. Detailed descriptions of the security mechanisms and cryptography behind those protocols are given in [BSI TR-03110].

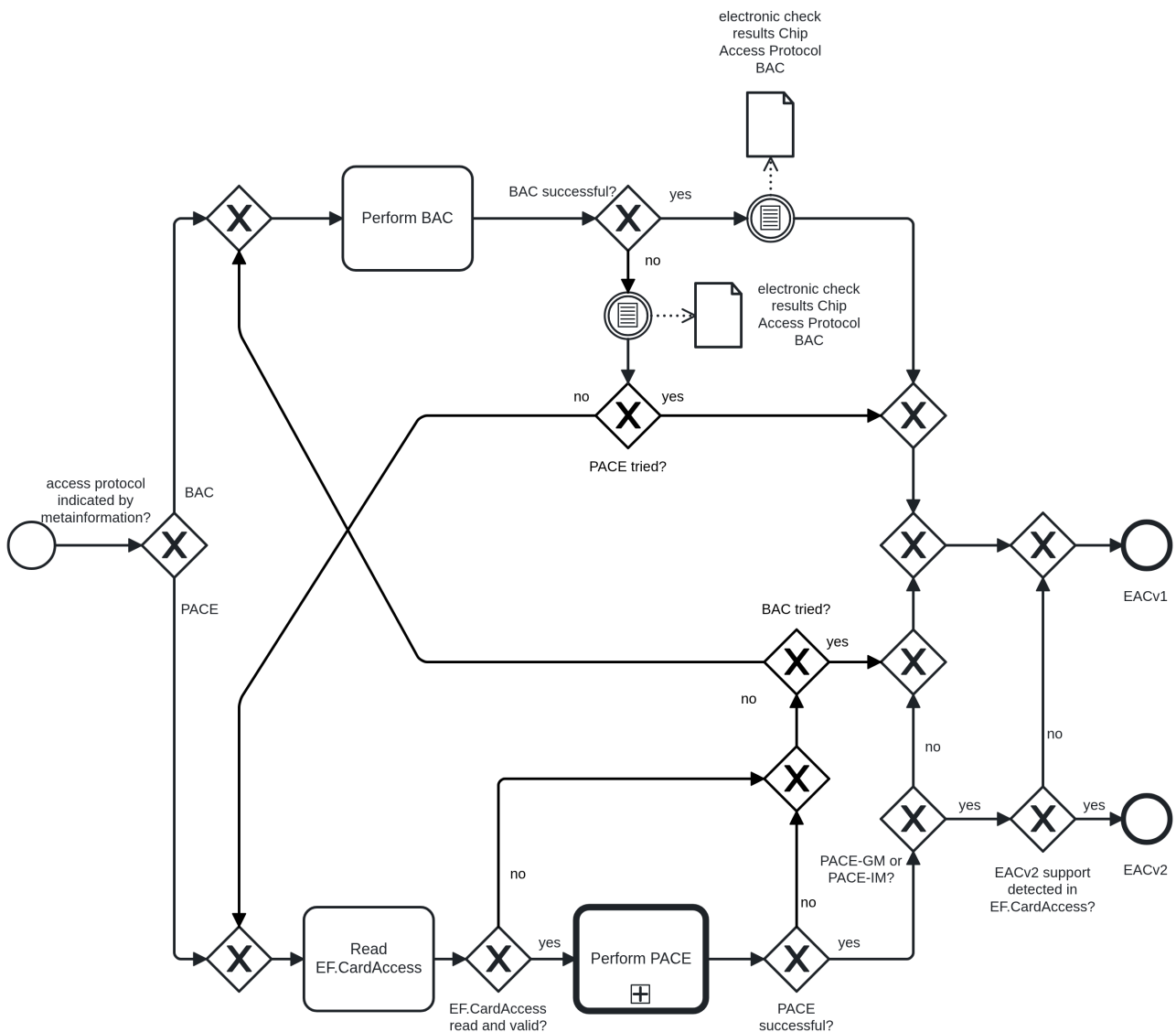


Figure 5.4. Subprocess "Chip Access"

Following the successful detection of an electronic chip, it is not yet known (to the IS) whether or not the chip supports BAC and/or PACE. In addition, it is possible that a document supports both PACE and BAC.

The IS SHALL provide a mechanism for the user to choose which protocol is to be tried first. The choice of the preferred protocol MAY be stored in a configuration file. If a chip access protocol is not supported or fails, the supported alternative protocol(s) SHALL be tried. If both BAC and PACE fail and/or are not supported, the IS SHALL continue the process and attempt to read the chip data over an insecure communication channel, if supported by the chip.

If PACE is to be performed, the IS SHALL determine which modes of the protocol are supported by the chip by reading and validating the `EF.CardAccess`. Subsequently, as shown in detail in ▶Figure 5.5, the IS SHALL attempt to perform the PACE protocol until successful or until all the available PACE modes have been tried. The IS MAY configure and control the order in which the available modes are attempted.

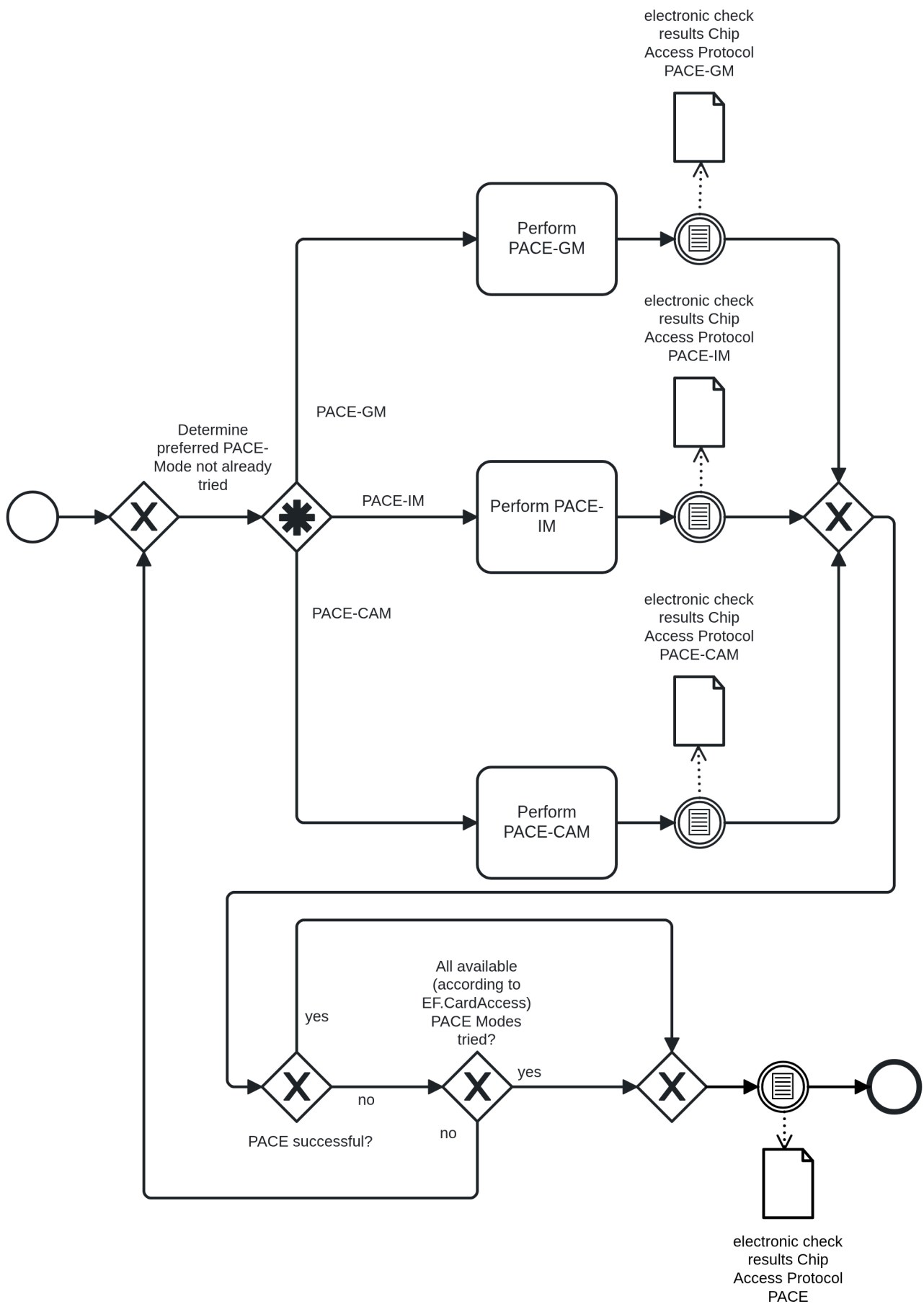


Figure 5.5. Subprocess "Perform PACE"

See ►Section 5.8.3 for an explanation of the possible check results of the "Chip Access" subprocess.

Based on the information in `EF.CardAccess` and/or the type, mode, and result of the performed chip access protocols, the IS SHALL determine the version of the EAC protocol which is to be used later on in the process, cf. overview in ►Figure 5.3 and details in ►Figure 5.4. For details about the different versions of EAC, see ►Section 5.8.10.2.

5.8.10.2 Extended access protocols

This section covers the "EAC" subprocesses of ►Figure 5.3. Detailed views of the EACv2 and EACv1 subprocesses are shown in ►Figure 5.6 and ►Figure 5.7, respectively.

If the information in `EF.CardAccess` indicates EACv2 to be conducted, the IS SHALL start the process with the TAv2 protocol. In case TAv2 has not been successful, the IS SHALL switch to EACv1 as a fallback.

The IS SHALL conduct the Passive Authentication (PA) and CA protocols to check the authenticity, integrity, and originality of the data stored on the chip. During passive authentication, the signature of the data stored on the document's chip is verified and traced back to the corresponding CSCA certificate. During chip authentication, a secure connection between the chip and the IS is established with a Diffie-Hellman key exchange, whereby the data and the chip are implicitly authenticated; this includes the detection of chip cloning. During EACv2, CAv2 or newer is performed, whereas during EACv1, CAv1 is performed. For more details on PA and CA, see [ICAO9303] and [BSI TR-03110].

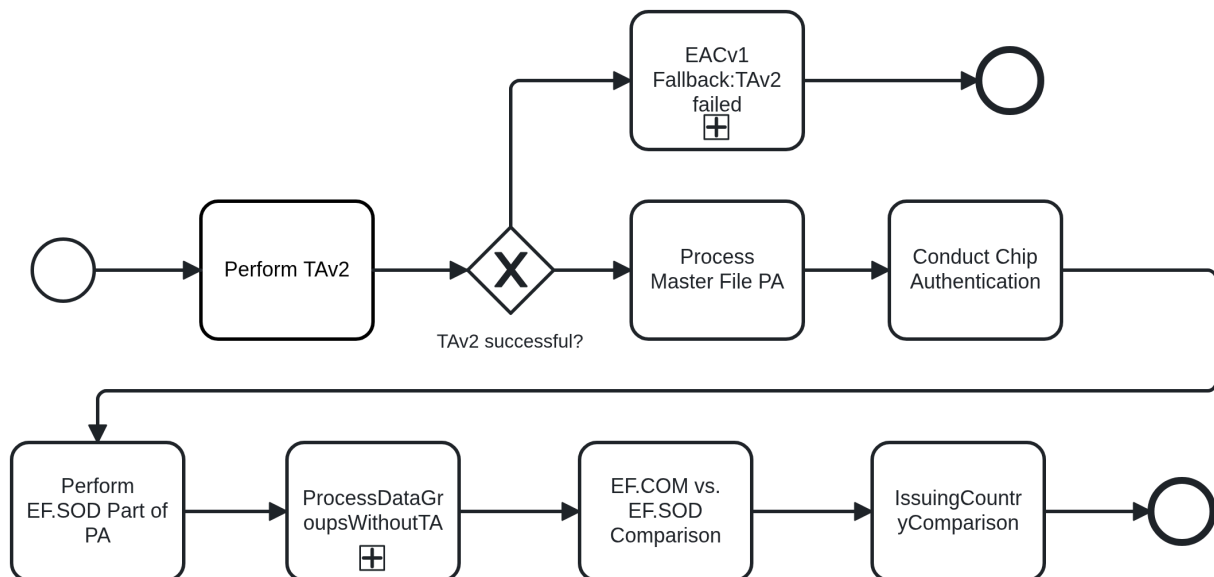


Figure 5.6. Subprocess "EACv2"

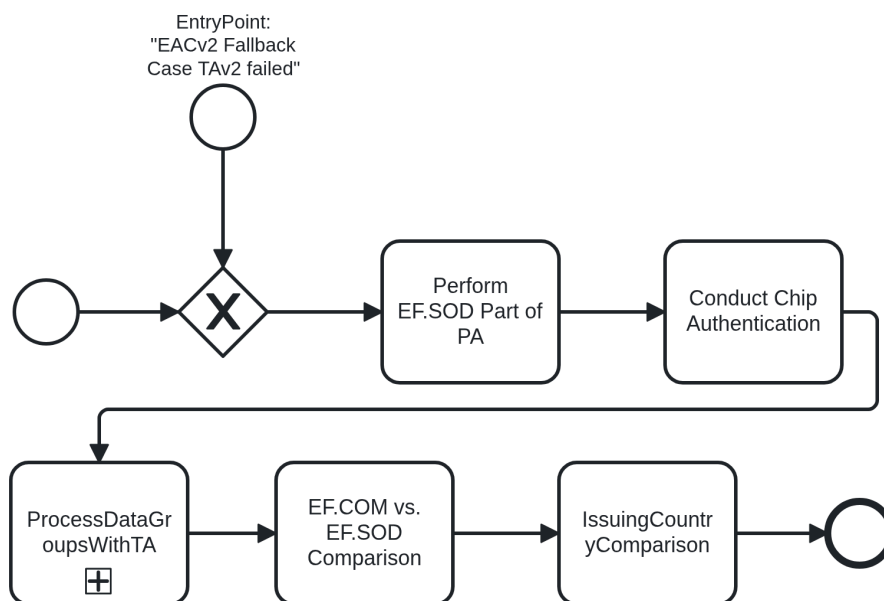


Figure 5.7. Subprocess "EACv1"

Following the passive authentication and chip authentication, the IS SHALL read the data from the chip, see figures ▶Figure 5.8 and ▶Figure 5.9 for an overview of this step in EACv1 and EACv2, respectively. First, the mandatory data groups, i.e. machine-readable zone information from DG1 and the facial biometric data from DG2 SHALL be read. While reading the biometric data of finger(s) from DG3 (if present) is OPTIONAL, a successful completion of TA is REQUIRED to do so. Reading the data from the remaining data groups (if present) is OPTIONAL. The IS SHOULD offer a configuration option specifying which non-mandatory data groups are to be read from the chip.

If both EF.COM and EF.SOD are present in the chip, the IS SHALL verify their consistency by checking whether or not all data groups listed in EF.SOD are included in EF.COM. Furthermore, the IS SHALL check whether or not the issuer of the DS certificate and the issuing state listed in EF.DG1 of the chip match according to an assignment table for 2-letter ISO and 3-letter ICAO country codes.

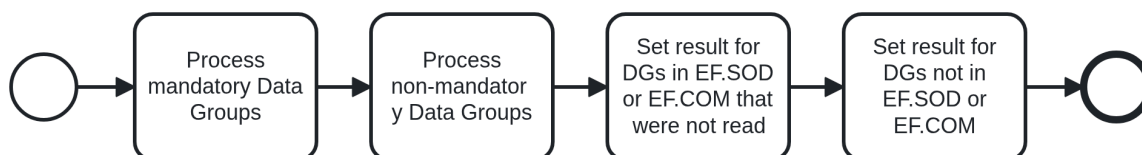


Figure 5.8. Subprocess "Process data groups without TA"

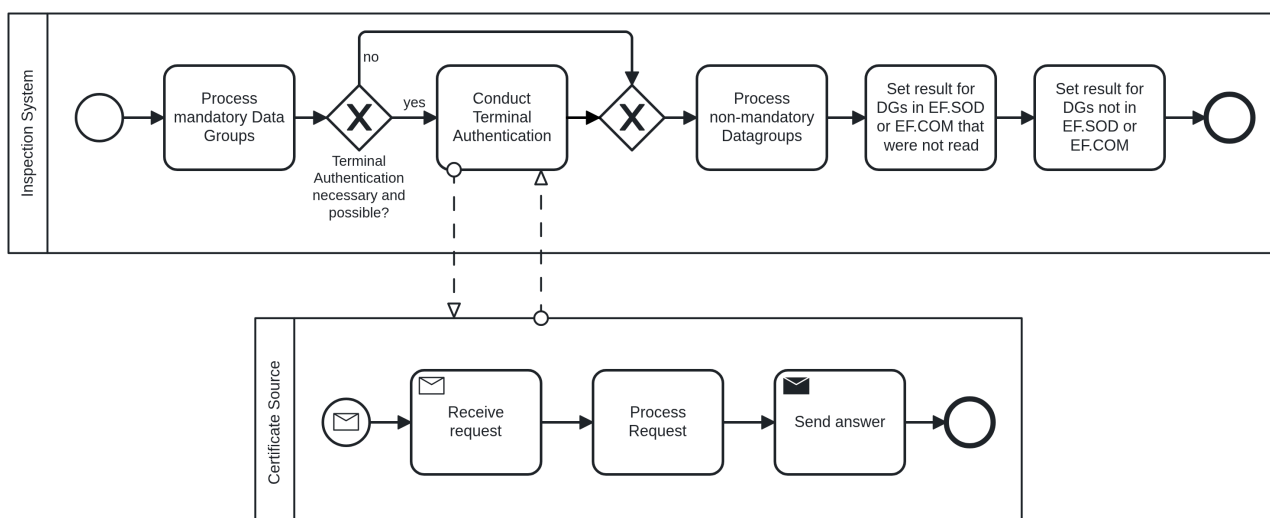


Figure 5.9. Subprocess "Process data groups with TA"

5.8.10.3 Result Aggregation

Once the protocols, checks, and data access described in the preceding subsections have been carried out, the IS SHALL conclude the process by aggregating the results of the conducted checks in accordance with ▶Section 3.8 and ▶Section 5.8.1. The IS SHALL log the check results and defects in accordance with ▶Chapter 6.

5.8.11 Additional chip access performing Terminal Authentication (TA)

In order to be able to read sensitive data from an EAC-protected document, a digital authorization token shall be provided to the RF-chip to gain access to EAC-protected data groups (e.g. EF.DG3). The requirements for the architecture of the PKI defined in chapter ▶Section 4.2 are a mandatory precondition for this check.

The result of TA SHALL be mapped according to ▶Table 5.116.

Result of the check	Description
successful	TA could be performed and the terminal is authorized to read sensitive data from the chip.
failed	The TA protocol failed.
undetermined	TA could not be performed due to missing certificates, e. g. the IS does not have all required certificates.
not supported	TA is not supported by the document.
aborted	Due to an error the checking process was aborted prior to its defined process flow.

Table 5.116 Final result of the access protocol Terminal Authentication

5.8.12 Integrity of chip characteristics

The previous sections described the security mechanisms of an eMRTD as described in [ICAO9303] and [BSI TR-03110]. Protocols such as PACE or Chip Authentication and Terminal Authentication, but also Passive Authentication, significantly increase the security of the chip and the whole document.

Normally, the inspection system reads the information provided by the eMRTD chip and then decides which protocols and algorithms to use (see figure ▶Figure 5.3 or section ▶Section 5.8.10). For example, the inspection system reads EF.CardAccess and then decides which of the offered PACE algorithms to perform. In this way, the inspection system trusts the chip in the eMRTD. If the chip is manipulated in such a way that it pretends to be a simple chip without any electronic security features or a chip with reduced security features (e.g. BAC instead of PACE), the inspection routines at border control will be bypassed. In this case, the inspection system could trust the chip and run no (or reduced) security protocols. This type of attack could be prevented by checking for the presence of the EF.SOD with semantically complete content. In general, however, it would be helpful to know which algorithms and protocols can be expected from an eMRTD.

To avoid such attacks, an additional reference database of chip information is helpful. Such a database is already widely used for optical security checks. With such a database, a chip can be checked for plausibility at the beginning of the electronic reading process. This database contains a 'fingerprint' of the chip with basic information about the chip. The inspection system can compare the expected data (database) with the data read from the chip (eMRTD). If there is a mismatch between the actual data and the expected data, the chip may be counterfeit.

This verification of the chip characteristics must not noticeably delay the process. In addition, the check must be robust enough not to compromise the security protocols.

A Forensic Analysis Information List (Forensic Analysis Information List (FAIL)) stores these chip characteristics, similar to the fault list, and an inspection system can use this list to verify that the eMRTD chip is performing as expected. This procedure increases the security during border control process.

5.8.13 Digital Travel Credentials (DTC)

Designed to support seamless travel at the airport, a DTC is intended to serve as a secure and globally interoperable digital representation of a physical eMRTD. A key feature of the DTC is that it allows border authorities to verify a digital representation of the eMRTD data prior to the traveler's arrival, confirming the integrity and authenticity of the data.

There are different types of DTC that consist of two distinct components. These components are the eMRTD DTC Virtual Component (DTC-VC) specified in [ICAODTC-VC] and the eMRTD DTC Physical Component (DTC-PC) specified in [ICAODTC-PC].

The DTC-VC is a digital representation of the eMRTD data that is cryptographically linked to the issuing authority (i.e., digitally signed by the issuer). This enables verifiers to check the digital signature to confirm the authenticity of the credentials and that the data has not been altered after issuance. The DTC-VC is a file that can be stored on any medium. It does not have any inherent access protection on its contents.

The DTC-PC is a kind of carrier for the DTC-VC and serves as a unique physical authenticator. Similar to the relationship between the eMRTD book and its embedded chip, the DTC-VC is cryptographically linked to its DTC-PC via digital signatures. This allows verifiers to confirm the association of a DTC-VC with its corresponding DTC-PC to further authenticate the DTC. The can be created on any token that supports the DTC specifications (smartphones, wearables, etc.).

A DTC can be implemented in four different ways, each offering different options for issuers and verifiers. It is important to note that these four DTC types are not hierarchical, nor do they represent evolving DTC solutions. Hence, all DTC types use the same technical specifications for the DTC-VC.

The different types of DTCs are shown in ▶Table 5.117.

Type	Description
eMRTD bound	An eMRTD serves as the physical authenticator. The DTC-VC contains the digital representation of the eMRTD data which is derived from the eMRTD chip. The Traveler needs to carry the eMRTD with them while traveling. This type of DTC is considered to be issued by the passport authority, because it is derived from an existing eMRTD.
eMRTD bound extended	This type is similar to the eMRTD bound DTCs with the following difference: Additional information (DTCOtherInfo), not present in the eMRTD, can be contained inside the DTC-VC. An example for this is a facial image of the holder in the format of [ISO/IEC-39794-5].
eMRTD-PC bound	In this type, there is a DTC-PC in addition to the underlying eMRTD. The issuing authority creates a DTC-PC on any token that supports the DTC requirements and cryptographically links it to the embedded DTC-VC. Hence, clients issued this type of DTC would hold an underlying physical eMRTD that has the same passport number as the DTC and can also serve as a physical authenticator for the DTC-VC.
PC-bound	This type is similar to eMRTD-PC bound with the following difference: There is only a DTC-PC and no underlying eMRTD. The passport authority issues the DTC-VC and creates the DTC-PC on a token similar to the eMRTD-PC bound DTC. An example for a use case are emergency travel documents.

Table 5.117 Different DTC Types

5.8.13.1 Reading a DTC

The process of reading a DTC depends on the type of the DTC.

The DTC types, eMRTD-PC bound and PC bound are alternative representations of information contained in the electronic chip embedded in the eMRTD. For instance, the same cryptographic protocols and the same data groups (including some new ones) are used in the context of the DTC. Only the underlying transmission protocols may be different, i.e. besides NFC, there may be new protocols such as WiFi Aware or BLE. On the logical layer, however, the same cryptographic protocols and the corresponding checks can be used as described in ▶Section 5.8.10.1. To put it another way, from a logging perspective, these two types of DTC can be handled similar to a common eMRTD.

The DTC types, eMRTD bound and eMRTD bound extended, are virtual representations of the chip's content. Therefore, the process at the border is different:

1. The traveller derives a DTC-VC from their eMRTD.
2. The traveller transmits the DTC-VC to the border control system.

3. To verify that the DTC belongs to the traveller, a facial image of the traveller is captured during the border control process.
4. A background system performs a 1:N search on the available facial images of all DTC-VC to find a match.
5. If a DTC-VC is found where the contained facial image matches with one in the background system, this DTC-VC is transmitted to the border control station.
6. PA is performed with the DTC-VC to assure the integrity of its data.
7. The border control system requests the traveller to present his eMRTD to verify the DTC-VC. For this, PACE establishes a secure channel to the eMRTD and CA is executed to ensure the authenticity of the eMRTD. Alternatively, PACE-CAM (a combination of PACE and CA) can be executed.
8. If all steps are successful, the border control system allows the traveller to pass.

In any case it is important to note that in contrast to the data groups on a chip, a DTC-VC is a file that can be modified by third parties. Hence, to assure that no data is modified, it is necessary to perform PA with the DTC-VC, i.e. all checks specified described in ▶Section 5.8.6 and ▶Section 5.8.7 must be performed for the DTC-VC. This way, the integrity of the data can be verified.

A DTC of type eMRTDbound extended with additional data stored in the virtual component, plays a special role in verification process. This is because all additional data in the virtual component must be signed by the issuing authority. Accordingly, during the border control process, this signature must also be checked to assure the integrity of the additional data.

5.9 Checking a digital seal component

The following sections describe how an IS SHALL inspect a digital seal component.

The specific checks to be performed depend on the application scenario and SHALL be defined in a profiling document.

If an IS detects multiple digital seal components, it SHALL inspect each one separately.

5.9.1 Overall result

The inspection system SHALL aggregate the overall result of checking a digital seal.

The result SHALL be mapped according to ▶Table 5.118.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.118 Final result of digital seal checks of an MRTD

5.9.1.1 Aggregation of digital seal check results

The digital seal checks are divided into partial checks. These checks are aggregated according to the specified rules from ▶Section 3.8. The associated return value is translated into traffic light colours and displayed as the result of the digital seal checks. The aggregation of these check results is shown in ▶Table 5.119. The left column of the table describes the check type and the right column the partial check of the digital seal checks.

Document Check Type	Partial Checks	Partial Checks	Partial Checks
Digital Seal Checks	▶Section 5.9.1.2: Check the readability of a digital seal ReadabilityCheck	▶Section 5.9.1.2.1: Check the presence of a digital seal PresenceDetectionCheck	
		▶Section 5.9.1.2.2: Check the visual to binary conversion VisualToBinaryConversionCheck	
		▶Section 5.9.1.2.3: Check the technology type conformance TechnologyTypeConformanceCheck	
	▶Section 5.9.1.3: Check the integrity of a digital seal IntegrityCheck	▶Section 5.9.1.3.1: Check the structure a digital seal StructureCheck	
		▶Section 5.9.1.3.2: Check the semantic requirements for digital seal SemanticCheck	
		▶Section 5.9.1.3.3: Check the MRZ data consistency MrzConsistencyCheck	Partial checks see ▶Table 5.4.
	▶Section 5.9.1.4: Check the issuer certificates of a digital seal IssuerCertificatesCheck	Partial checks see ▶Table 5.29.	
	▶Section 5.9.1.5: Check the authenticity of a digital seal AuthenticityCheck	▶Section 5.9.1.5.1: Check the signature value SignatureValueCheck	
		▶Section 5.9.1.5.2: Check the digest DigestCheck	
	▶Section 5.9.1.6: Check the issuing country of a digital seal IssuingCountryCheck	Partial checks see ▶Table 5.24.	

Table 5.119 Aggregation of combined document check results

5.9.1.2 Check the readability of a digital seal

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ▶Table 5.120.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.

Result	Description
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.120 Result of checking the readability of a digital seal

5.9.1.2.1 Check the presence of a digital seal

The IS SHALL compare the expected and actual presence of a digital seal (e.g. a 2D barcode) on the document under inspection.

The result of this check SHALL be mapped according to ►Table 5.121.

Result of the check	Description
successful	At least one digital seal was detected on the document.
failed	A digital seal was expected on the document but none could be detected.
undetermined	This result SHALL NOT occur here.
not supported	No digital seal was expected on the document and none was detected on the document.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.121 Result of checking the presence of a digital seal

5.9.1.2.2 Check the visual to binary conversion

The IS SHALL convert the visual representation (digital image) of the digital seal to a binary representation (encoded bytes) of its contents.

The result of this check SHALL be mapped according to ►Table 5.122.

Result of the check	Description
successful	The visual representation was converted to binary data.
failed	The visual representation was not converted to binary data.
undetermined	This result SHALL NOT occur here.
not supported	The technology used for the digital seal is not supported by the IS.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.122 Result of checking the visual to binary conversion

5.9.1.2.3 Check the technology type conformance

The IS SHALL check whether the expected and actual technology type of the digital seal match.

The result of this check SHALL be mapped according to ►Table 5.123.

Result of the check	Description
successful	The expected and actual technology type of the digital seal match.
failed	The expected and actual technology type of the digital seal do not match.
undetermined	This result SHALL NOT occur here.

Result of the check	Description
not supported	There is no knowledge about the expected technology type available to the IS.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.123 Result of checking the technology type conformance

5.9.1.3 Check the integrity of a digital seal

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The aggregation result SHALL be mapped according to ►Table 5.124.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.124 Result of checking the integrity of a digital seal

5.9.1.3.1 Check the structure a digital seal

The IS SHALL check whether the expected structure of the data contained in the digital seal matches the actual structure.

The result of this check SHALL be mapped according to ►Table 5.125.

Result of the check	Description
successful	The actual structure and the expected structure match.
failed	The actual structure and the expected structure do not match.
undetermined	This result SHALL NOT occur here.
not supported	There is no knowledge about the expected structure available.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.125 Result of checking the presence of a digital seal

5.9.1.3.2 Check the semantic requirements for digital seal

The IS SHALL check that the data contained in the digital seal conforms to the semantic requirements.

The result of this check SHALL be mapped according to ►Table 5.126.

Result of the check	Description
successful	The data contained in the digital seal conforms to the semantic requirements. Note: This does include a deliberately empty set of semantic requirements for a specific digital seal.
failed	The data contained in the digital seal does not conform to the semantic requirements. E.g. for a seal that is expected to be digitally signed there is neither a certificate nor a certificate reference present.
undetermined	This result SHALL NOT occur here.

Result of the check	Description
not supported	There is no knowledge about the semantic requirements available. Note: This DOES NOT include deliberately specifying an empty set of semantic requirements (see result successful).
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.126 Result of checking the semantic of a digital seal

5.9.1.3.3 Check the MRZ data consistency

Digital seals can contain MRZ data.

The IS SHALL perform all checks given in ▶Section 5.6.1: Check the MRZ data for the specific application scenario.

5.9.1.4 Check the issuer certificates of a digital seal

The IS SHALL perform the checks given in ▶Section 5.6.3.2 for the certificate contained in (or referenced by) a digital seal.

5.9.1.5 Check the authenticity of a digital seal

The IS SHALL aggregate the results of all checks prescribed in the subsections for the application scenario.

The result of this check SHALL be mapped according to ▶Table 5.127.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.127 Result of checking the authenticity of a digital seal

5.9.1.5.1 Check the signature value

The IS SHALL check the cryptographic signature value of the seal data.

The result of this check SHALL be mapped according to ▶Table 5.128.

Result of the check	Description
successful	The seal's signature is cryptographically correct.
failed	The seal's signature is cryptographically not correct.
undetermined	Information is missing for verifying the signature (e.g. the signature algorithm is unknown).
not supported	The digital seal does not contain a signature.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.128 Result of checking the signature value of the seal data

5.9.1.5.2 Check the digest

The IS SHALL compare the digest contained inside the digital seal with the digest calculated by the IS.

The result of this check SHALL be mapped according to ►Table 5.129.

Result of the check	Description
successful	The two comparands are equal.
failed	The two comparands are not equal.
undetermined	The comparison was not carried out, e.g. because the hash algorithm used is not known to the IS.
not supported	The digital seal does not contain a digest.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.129 Result of checking the digest

5.9.1.6 Check the issuing country of a digital seal

The IS SHALL perform the checks given in ►Section 5.6.2.2 for the certificate contained in (or referenced by) a digital seal.

5.10 Combined checks

The following sections describe how an IS SHALL perform combined checks that concern multiple document components.

The specific combined checks to perform depend on the application scenario and SHALL be defined in a profiling document.

The inspection system SHALL aggregate the overall result of performing combined checks.

The result SHALL be mapped according to ►Table 5.130.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.130 Final result of combined checks of an MRTD

5.10.1 Aggregation of combined check results

The combined checks are divided into partial checks. These checks are aggregated according to the specified rules from ►Section 3.8. The associated return value is translated into traffic light colours and displayed as the result of the combined checks. The aggregation of these check results is shown in ►Table 5.131. The left column of the table describes the check type and the right column the partial check of the combined checks.

Document Check Type	Partial Checks	Partial Checks
Combined Check	►Section 5.10.2: Check the document's date of expiry	
Checks	DateOfExpiryCheck	

Document Check Type	Partial Checks	Partial Checks
	▶Section 5.10.3: Check the document's date of expiry against signer certificate validity MrzVsSignerExpiryCheck	
	▶Section 5.10.4: Check the consistency between MRZ and EF.DG1 MrzVsDg1Check	▶Section 5.6.4: Check the consistent representation of one component ComponentCheck
	▶Section 5.10.5: Check the consistency between MRZ and Digital Seal MrzVsSealCheck	▶Section 5.6.4: Check the consistent representation of one component ComponentCheck
	▶Section 5.10.6: Check the consistency between VIZ and Digital Seal VizVsSealCheck	▶Section 5.6.4: Check the consistent representation of one component ComponentCheck
	▶Section 5.10.7: Check the consistency between EF.DG2 and the VIZ facial image Dg2VsVizFacialImageCheck	
	▶Section 5.10.8: Check the consistency between the passport number from VIZ and Digital Seal VizVsSealPassportNumberCheck	
	▶Section 5.10.9: Check the presence of a chip ChipPresenceCheck	

Table 5.131 Aggregation of combined document check results

5.10.2 Check the document's date of expiry

The IS SHALL compare the date of expiry of the document with the date of inspection to determine whether the document is still valid.

The result of this check SHALL be mapped according to ▶Table 5.132.

In this process, a single date of expiry needs to be determined. For this, validated dates of expiry based on digitally signed data (e.g. EF.DG1) SHALL be given preference. Further, if no such validated expiration date is available, but the date of expiry can be derived from the VIZ, the expiration date SHALL be taken from the VIZ. Otherwise, the date of expiry from the optical MRZ SHALL be employed. The date of expiry used for this check SHALL be the same that is used for the check ▶Section 5.10.3.

If the date of expiry is obtained from MRZ data, the year of expiration is only represented by the last two digits and the following rules SHALL be applied to calculate the expiration date (year):

- By means of the valid times of the certificates stored in the electronic component or Digital Seal, the actual expiration date can be precisely calculated. Therefore, the validity information of the DS certificate SHALL be used, specifically, choose the smallest natural number as the first two digits of the document's expiration date (century digits) such that the expiration date is after the start of validity of the DS certificate.
- Otherwise, if no validity information of the DS certificate is available, the following rule SHALL apply:
 - If the two digits of the year from the MRZ data are between the current year and 15 years ³ in the future, the actual year SHALL be added in such a manner that the complete expiration date is in the future (e.g. extracted expiration date from the MRZ data = 150401, current date = 02.05.2012, actual expiration date = 01.04.2015).

³ For previously known documents with a longer period of validity, another value MAY be assumed here, e.g. the Russian national passport or the Romanian ID card. In addition to this, there might exist deviating periods during which the document is accepted for crossing the border, although it has expired, e.g. the French passport.

- If the two digits of the year from the MRZ data not between the current year and 15 years in the future, it SHALL be assumed that the actual expiration date is in the past (e.g. extracted expiration date from the MRZ data= 110302, current date = 02.05.2012, actual expiration date = 02.03.2011).

The system MAY perform an alternative check of validity if there is knowledge about model specific expiration dates.

The result of this check SHALL be mapped according to ►Table 5.132.

Result of the check	Description
successful	The date of expiry of the document is equal to or after the date of inspection.
failed	The date of expiry of the document is before the date of inspection.
undetermined	The date of expiry of the document cannot be extracted from the document.
not supported	This value cannot occur here.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.132 Result of checking the dates of expiry

5.10.3 Check the document's date of expiry against signer certificate validity

The IS SHALL check whether the date of expiry of the document is consistent with the end of validity of a (document, seal) signer certificate. The date of expiry of the document used for this check SHALL be the same that is used for the check ►Section 5.10.2. This check SHALL be separately performed for all signer certificates that are used during the inspection of the document.

The result of this check SHALL be mapped according to ►Table 5.133.

Result of the check	Description
successful	The date of expiry is before the end of validity of a specific signer certificate.
failed	The date of expiry is after the end of validity of a specific signer certificate or the signer certificate has no end of validity.
undetermined	This value cannot occur here.
not supported	There were no signer certificates used during the inspection or the document does not contain a date of expiry.
not supported	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.133 Result of checking the dates of expiry of the MRZ sources and a signer certificate

5.10.4 Check the consistency between MRZ and EF.DG1

The IS SHALL check whether the optical MRZ data is consistent with EF.DG1 by executing a component check as described in ►Section 5.6.4 for each individual element (first name, last name, date of birth, sex, date of expiry, optional data, etc.) expected to be included in both sources for the identified document model.

x

All elements SHALL match each other exactly with the following exceptions, deviating from [ICAO9303]:

- The date of birth, if not all 6 characters have been filled with numbers, but with so called fill characters (“<”). The exception only refers to these fill characters.
- If only the optional data is not consistent, the result SHALL NOT be failed.

The result of this check SHALL be mapped according to ►Table 5.134.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.

Result	Description
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.134 Result of checking the consistency between MRZ and EF.DG1

5.10.5 Check the consistency between MRZ and Digital Seal

The IS SHALL check whether the MRZ data printed on the document is consistent with the digital seal by executing a component check as described in ►Section 5.6.4 for each individual element (first name, last name, date of birth, sex, date of expiry, optional data, etc.) expected to be included in both sources for the identified document model.

Please note that the digital seal may only contain a part of the MRZ data. Therefore, this check requires consistency, not an exact match.

The result of this check SHALL be mapped according to ►Table 5.135.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.135 Result of checking the consistency between MRZ and the digital seal

5.10.6 Check the consistency between VIZ and Digital Seal

The IS SHALL check whether the VIZ data is consistent with the digital seal by executing a component check as described in ►Section 5.6.4 for each individual element (first name, last name, date of birth, sex, date of expiry, optional data, etc.) expected to be included in both sources for the identified document model.

Please note that the digital seal may only contain a part of the VIZ data and vice versa. Therefore, this check requires consistency, not an exact match.

The result of this check SHALL be mapped according to ►Table 5.136.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.136 Result of checking the consistency between VIZ and digital seal

5.10.7 Check the consistency between EF.DG2 and the VIZ facial image

The IS SHALL check whether the facial image contained in EF.DG2 and the printed facial image are consistent.

The result of this check SHALL be mapped according to ►Table 5.137.

Result of the check	Description
successful	The facial image in EF.DG2 and the printed facial image are consistent.
failed	The facial image in EF.DG2 and the printed facial image are not consistent.
undetermined	This value cannot occur here.
not supported	The document does not contain a facial image or a EF.DG2.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.137 Result of checking the consistency between EF.DG2 and the VIZ facial image

5.10.8 Check the consistency between the passport number from VIZ and Digital Seal

The IS SHALL check whether the VIZ passport number is consistent with the one in the digital seal.

The result of this check SHALL be mapped according to ►Table 5.138.

Result of the check	Description
successful	The passport numbers between VIZ and digital seal are consistent.
failed	The passport numbers between VIZ and digital seal are not consistent.
undetermined	This value cannot occur here.
not supported	The document does not contain a passport number in the VIZ or the seal or at least one of the components (VIZ, seal) does not exist.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.138 Result of checking the consistency between the passport number from VIZ and digital seal

5.10.9 Check the presence of a chip

The IS SHALL check if the document contains a chip based on visual cues or databases.

The result of this check SHALL be mapped according to ►Table 5.139.

Result of the check	Description
successful	The chip presence is as expected.
failed	The chip presence is not as expected.
undetermined	This value cannot occur here.
not supported	There are no visual cues and no database present.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.139 Result of checking the presence of a chip

5.11 Cross document checks

The following sections describe how an IS SHALL check the consistency of two separate documents that belong to the same person. This type of check is called a cross document check.

A cross document check is performed when the inspection system is used to perform document checks on at least two different related documents. The inspection system SHALL perform a cross document check for all pairs of related documents for which it is able to perform it.

Examples for related documents are:

- passport
- residence permit
- visa
- identity card

In general, if available, data that was successfully verified via a digital signature (i.e. data from electronic components and digital seals) SHOULD be used for cross document checks.

5.11.1 Overall result

The specific cross document checks to perform depend on the application scenario and SHALL be defined in a profiling document.

The inspection system SHALL aggregate the overall result of performing combined checks.

The result SHALL be mapped according to ►Table 5.140.

Result	Description
successful	The result SHALL be set to successful if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed, aborted or undetermined and at least one had the result successful.
failed	The result SHALL be set to failed if the process of aggregating the result was not aborted and at least one of the checks prescribed in the subsections for the application scenario had the result failed.
undetermined	The result SHALL be set to undetermined if the process of aggregating the result was not aborted and none of the checks prescribed in the subsections for the application scenario had the result failed or aborted and at least one had the result undetermined.
not supported	The result SHALL be set to not supported if the process of aggregating the result was not aborted and all of the checks prescribed in the subsections for the application scenario had the result not supported.
aborted	The result SHALL be set to aborted if the process of aggregating the result was aborted or none of the checks prescribed in the subsections for the application scenario had the result failed and at least one had the result aborted.

Table 5.140 Overall result of cross checks of an MRTD

5.11.2 Cross document check aggregation

A cross document check is divided into partial checks. These partial checks are aggregated according to the specified rules from ▶Section 3.8. The aggregation of these check results is shown in ▶Table 5.141. The left column of the table describes the check type and the right column the partial check of the cross checks.

Document Check Type	Partial Checks
Cross Check Checks	▶Section 5.11.3: MRZ data consistency check MrzDataCheck
	▶Section 5.11.4: Passport number consistency check PassportNumberCheck
	▶Section 5.11.5: Date of expiry consistency check ExpiryCheck

Table 5.141 Aggregation of cross document check results

5.11.3 MRZ data consistency check

The IS SHALL check whether the MRZ data is consistent between the two documents with respect to the mandatory data. For this, the IS SHALL use the MRZ data determined for the respective document.

The result of this check SHALL be mapped according to ▶Table 5.142

Result of the check	Description
successful	The MRZ data is consistent between the two documents.
failed	The MRZ data is not consistent between the two documents.
undetermined	This value cannot occur here.
not supported	At least one document does not contain MRZ data.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.142 Result of checking the consistency of the MRZ data

5.11.4 Passport number consistency check

The IS SHALL check whether the passport number is consistent between the two documents (usually a passport and a visa).

The result of this check SHALL be mapped according to ▶Table 5.143

Result of the check	Description
successful	The passport number is consistent between the two documents.
failed	The passport number is not consistent between the two documents.
undetermined	This value cannot occur here.
not supported	At least one document does not contain a passport number.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.143 Result of checking the consistency of the passport number

5.11.5 Date of expiry consistency check

The IS SHALL check whether the date of expiry is consistent between the two documents.

The result of this check SHALL be mapped according to ▶Table 5.144

Result of the check	Description
successful	The date of expiry is consistent between the two documents.
failed	The date of expiry is not consistent between the two documents.

Result of the check	Description
undetermined	This value cannot occur here.
not supported	At least one document does not contain a date of expiry.
aborted	Due to an error, the checking process was aborted prior to its defined process flow.

Table 5.144 Result of checking the consistency of the dates of expiry

5.12 Handling and interpretation of defects

During production of MRTDs errors may occur, both in the optical/physical parts and in the electronic parts. This section defines a procedure in the event of defects that may be ignored for the final result. If a defect was detected there MAY be a hint provided to the operator as to this fact.

5.12.1 Optical defects

There are documents issued that do not comply to requirements specified in [ICAO9303]. In such cases a check will fail, although the document was officially issued by a state. If such defects can be identified it is RECOMMENDED to provide additional information about the optical defect to the operator, or to log such information for the operational monitoring.

5.12.1.1 Specimen

There are specimen documents in the field that correspond to authentic original documents. Such documents usually contain all security-related features of a MRTD and could be successfully checked. Such documents contain specimen labels that help to identify them as a demonstration or fake document. If possible, such documents SHOULD be identified and marked as specimen.

5.12.1.2 Country identifier

E. g. visas are sometimes issued with an incorrect country naming scheme. For example, instead of D for Germany some countries issue DEU, GER or DE. But there are also other examples for a wrong country encoding. If it is known that a wrong country code was used, a defect SHOULD be specified and can be corrected for a further check.

5.12.1.3 MRZ format

In some passports and visa stickers the MRZ-format does not comply to [ICAO9303]. Regarding the individual defect, the MRZ might be corrected. If it is not possible to do so, checks relating on a correct MRZ-format SHOULD be skipped for such documents.

5.12.1.4 MRZ font

In some passports and visa stickers, a non-OCR-B font was used to print the MRZ. If it is known that an issuing country uses a wrong font it is RECOMMENDED to discard any optical checks relating on the exact OCR-B-font. In case a digitized MRZ is editable, the defect SHOULD be specified and can be corrected.

5.12.1.5 MRZ check digit

Some passports and visa stickers contain wrong MRZ check digit values, because the issuing state missed the algorithm or set a default value, regardless of the individual document holder's MRZ data. If it is known that the MRZ check digit is malformed and will fail, checks relating the MRZ check digit SHOULD be skipped or the digitized MRZ SHOULD be corrected if possible.

5.12.1.6 MRZ non-IR-readable

In some documents the MRZ was printed using IR transparent ink., if it is known, that the MRZ was printed using IR transparent ink this defect can be corrected by reading the MRZ using the VI image.

5.12.2 Electronic defects

There are documents where information in the chip is malformed or not compliant to standards. In such cases an electronic check will fail if there is no mechanism to correct or ignore the malformed information.

Defect Lists define such errors and how to heal or move on. They SHALL only be processed according to the formats and protocols defined in [BSI TR-03129-2] in order to signal production errors in documents as well as revocations of already issued and used certificates. This section defines the procedure of the Inspection System to be implemented in the event of defects. Please note that there still exist defects that cannot be corrected. This technical guideline does not specify defects for the Electronic Identity Document (eID) application.

5.12.2.1 Authentication defects

5.12.2.1.1 Document Signer Certificate Revoked

The defect `Document Signer Certificate Revoked` SHOULD be used for DS certificates contained in the electronic component or digital seals.

This defect is replaced by the more general defect `Signer Certificate Revoked` ([BSI TR-03129-2]) starting BSI TR-03129-2 version 1.4.2.

The status code of the defect affects the result of checking the trust status (cf. ▶Section 5.6.3.2.4) which is returned to the Inspection Application as a response. The result of the check SHALL depend on the defect according to ▶Table 5.145.

5.12.2.1.2 Signer Certificate Revoked

If the defect `Signer Certificate Revoked` (see [BSI TR-03129-2]) is detected for a certificate, the IS SHALL perform the checks on that certificate. However, the status code of the defect affects the result of checking the trust status (cf. ▶Section 5.6.3.2.4) which is returned to the Inspection Application as a response. The result of the check SHALL depend on the defect according to ▶Table 5.145.

Status code	Description of the status code	Result of checking the trust status of a certificate
0	no further details available	failed (i.e not trusted)
1	revocation is checked	undetermined
2	the certificate is used for test purposes	failed (i.e not trusted)
3	revoked by the issuer	failed (i.e not trusted)
4	revoked by the Defect List Signer	failed (i.e not trusted)
5	error known	undetermined
≥ 32	proprietary status codes	undetermined

Table 5.145 Trust status assignments for status codes of Defect Lists

5.12.2.1.3 Document Signer Certificate Malformed

If the defect `Document Signer Certificate Malformed` ([BSI TR-03129-2]) is detected for a DS certificate, the DS certificate given to the IS is no longer used. The signature check SHALL be set to valid without performing a check. The validity information of the response SHALL be taken from the replacement certificate returned for this defect. The trust status of the response of the IS SHALL be set to trusted.

A defect of this type has the effect that all related cryptographic checks SHALL be performed using the replacement certificate transmitted by the IS. The validity check SHALL be performed with the information provided through the replacement certificate and the cross check of the issuing country from the MRZ against the issuing country as specified in the DS certificate SHALL also be performed using the provided replacement certificate (see checks in sections ▶Section 5.6.3.2, ▶Section 5.6.3.2.2, ▶Section 5.6.2.1, ▶Section 5.10.2).

5.12.2.1.4 Chip Authentication Private Keys Compromised

If the defect `Chip Authentication Private Keys Compromised` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application.

The occurrence of such a defect SHALL make the Inspection Application assess the chip authenticity check (CA check) as failed.

5.12.2.1.5 Active Authentication Private Keys Compromised

If the defect `Active Authentication Private Keys Compromised` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application.

The occurrence of such a defect SHALL make the Inspection Application assess the chip authenticity check (active authentication) as failed.

5.12.2.1.6 Authentication Protocol Failure

If the defect `Authentication Protocol Failure` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application. The check SHALL be performed and if its check result is failed it SHALL be set to undetermined.

5.12.2.1.7 Validity Period Incorrect

If the defect `Validity Period Incorrect` ([BSI TR-03129-2]) is detected, the IS SHALL still perform checks on electronic documents and can also check the certificate chain with regard to the cryptographic protocols. The Inspection Application SHALL classify the result as undetermined and SHALL NOT be classified as failed.

5.12.2.2 Application defects

5.12.2.2.1 Data Group Malformed

If the defect `Data Group Malformed` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application.

The Inspection Application SHOULD provide information on improperly coded data groups.

5.12.2.2.2 Document Security Object Malformed

If the defect `Document Security Object Malformed` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application. If the parameter `EF.SOD` fault status is transmitted, this information can also be transmitted to the Inspection Application.

In the event of a failed signature verification regarding the security object (`EF.SOD`), the Inspection Application SHALL classify the result as undetermined and NOT as failed. In this case, the result of the verification of the security object (`EF.SOD`) SHALL be set to undetermined.

5.12.2.2.3 EF.COM and EF.SOD Discrepancy

If the defect `EF.COM and EF.SOD Discrepancy` ([BSI TR-03129-2]) is detected, the Inspection Application SHALL set the result of the partial check to successful for the data groups that can be verified using `EF.SOD` and, in all other cases, the result of the verification SHALL be set to undetermined.

5.12.2.2.4 Wrong Signer Identifier

If the defect `Wrong Signer Identifier` ([BSI TR-03129-2]) is detected, the Inspection Application did not find a corresponding DS in the security object for the given Signer Identifier. If the Inspection System replies with this defect and provides a DS certificate for the Signer Identifier, the returned DS certificate SHALL be used by the Inspection Application to obtain the trust status and to perform further electronic checks.

5.12.2.2.5 Issuing Country Defect

If the defect `Issuing Country Defect` ([BSI TR-03129-2]) is detected, the referenced issuing country was wrong, e. g. ICAO 3-letter code instead of ICAO 2-letter code, or a missing country code, or the issuer used a wrong issuing country code. This defect provides a country value which SHALL then be used to perform the electronic checks.

5.12.2.3 Document defects

5.12.2.3.1 Card Security Object Malformed

If the defect `Card Security Object Malformed` ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application.

In the event of a failed signature verification of the security object (EF.CardSecurity), the Inspection Application SHALL classify the result as undetermined and NOT as failed. In this case the result of the verification of the security object (EF.CardSecurity) SHALL be set to undetermined.

5.12.2.3.2 Chip Security Object Malformed

If the defect Chip Security Object Malformed ([BSI TR-03129-2]) is detected, the IS SHALL pass this defect to the Inspection Application.

In the event of a failed signature verification of the security object (EF.ChipSecurity), the Inspection Application SHALL classify the result as undetermined and not as failed. In this case the result of the verification of the security object (EF.ChipSecurity) SHALL be set to undetermined.

5.12.2.3.3 Requirement for switching off the chip (power down)

There exist chips that require a power down in between processing steps, meaning switching off the RF-based power supply. This Technical Guideline does not define or make any assumptions on switching/powering off the chip.

5.12.2.3.4 Document Signer Certificate incorrectly encoded or malformed

If the defect Document Signer Certificate incorrectly encoded or malformed ([BSI TR-03129-2]) is detected, this SHALL lead accordingly to the result undetermined.

6 Logging scheme for operational monitoring

6.1 Use cases

This chapter specifies a logging scheme, which makes it possible to document all technical activities performed during an inspection process. Such a logging scheme SHALL be used to measure the quality of the inspection process across different systems, regardless of the used hard- or software. This allows the implementation of operational monitoring:

- To optimize the process of machine assisted checks on eMRTDs,
- To adapt to new circumstances, and
- To ensure that reliability, quality, transparency and traceability is given.

6.2 XML schemas

The logging scheme is based on a transactional logging format which collects check results from the different application domains.

The logging format for this TR is defined in XML schemas. These schemas are self-documented and aligned to the structure of this guideline. The requirements included in the schema documentation SHALL be implemented by the vendor. For each check routine as defined in this guideline, a corresponding XML logging element exists in the schema. Additional elements are introduced to store context information of the inspection process (e. g. location and time, hardware and software component information).

►Table 6.1 lists the application domains and the corresponding schema files.

Please note the following:

- The definition of the schema of the background check node has been moved to the [BSI TR-03156].
- The biometric node that was present in the previous versions of this TR, SHALL be replaced by a separate [BSI TR-03121] log.
- It is RECOMMENDED to use the [BSI TR-03156] with integrated document check and cross document check nodes instead of the transaction layer, i.e. ct node.

Application domain	Schema file	Namespaces	Description
Transaction layer	ct-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/ct/5	system level information and wrapper for document checks, cross document checks and application specific checks
	bc4v4.xsd	http://tr03156.bsi.bund.de/bc/4	definition of background checks imported from TR-03156, see notes above.
Document checks	dc-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/dc/5	overall schema
	dco-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/dco/5	optical checks
	dce-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/dce/5	electronic checks
	dcc-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/dcc/5	checks that combine optical, electronic and/or digital seal information
	dcs-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/dcs/5	digital seal checks
Cross document checks	xdc-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/xdc/5	checks that compare different documents

Application domain	Schema file	Namespaces	Description
Application specific checks	-	-	Freely definable by the application scenario. A dedicated XML Schema Definition (XSD) schema SHALL exist for this data and SHALL be made available to the operator.
Base	ba-se-v5_0_1.xsd	http://trdoccheck.bsi.bund.de/base/5	types used by several different log schema files
Shared	log_hldc_shared-v5_0_1.xsd	-	types shared between log schema and hldc schema
Data protection	bsi-tr-additionaldata-ta-1v1.xsd	http://bsi-tr-additionaldata.bsi.bund.de/tradd/1	types used for data protection by different log types and log schema files

Table 6.1 Application domains and corresponding schema files

6.3 Logging profiling

Any profiling based on the requirements of this guideline SHOULD provide one or more dedicated logging configurations to indicate which elements are to be logged under which circumstances.

The following elements SHALL be logged in any case (i.e. if they are present after the process has finished):

- Transaction identifiers,
- Component version information,
- Timing information,
- All check results,
- Document data that is not related to a person,
- Statistical data about users that is not related to persons.

Logging of additional elements is application specific and needs to be defined by the system operator. Specifying the exact requirements SHOULD therefore be conducted together with the responsible Data Protection Officer.

Possible examples for use cases are:

- logging of document information in the case of suspicion of document fraud for deferred analysis,
- logging of document data forensic application.

6.4 Sequence of events

A log of a document inspection process can contain various timestamps and durations for different aspects of the inspection process, for example the optical or electronic checks. For this, the following sections define and describe all timestamps and durations. The last section (►Section 6.4.8) considers these timing aspects with regard to the performance of an IS.

6.4.1 Check transaction

The timestamps and durations regarding a check transaction are given in ►Table 6.2.

Short name	XPath	Description
CT Start Time	<code>/ct:CheckTransaction/ct:TimeMeasurement/ct:Overall/base:StartTime</code>	A transaction can encompass document checks and cross document checks. A transaction starts with a workflow.

Short name	XPath	Description
CT End Time	/ct:CheckTransaction/ct:TimeMeasurement/ct:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the transaction. CT End Time > all of [DC End Time, XDC End Time]
CT Duration	/ct:CheckTransaction/ct:TimeMeasurement/ct:Overall/base:Duration	CT Duration := CT End Time - CT Start Time

Table 6.2 Timestamps and durations to be logged by an inspection system - Check Transaction

6.4.2 Document check

The timestamps and durations regarding a document check are given in ►Table 6.3.

Short name	Relative XPath	Description
DC Start Time	/dc:DocumentCheck/dc:TimeMeasurement/dc:Overall/base:StartTime	The point in time when a document is presented to the document reader. This means that the reader tries to detect a document optically or electronically. The actual start time is given by the start time of the workflow.
DC End Time	/dc:DocumentCheck/dc:TimeMeasurement/dc:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the document check.
DC Duration	/dc:DocumentCheck/dc:TimeMeasurement/dc:Overall/base:Duration	DC Duration := DC End Time - DC Start Time
DC Document Model Selection Start Time	/dc:DocumentCheck/dc:TimeMeasurement/dc:DocumentModelSelection/base:StartTime	The point in time when the IS starts the process to requests the model selection from the user.
DC End Document Model Selection Time	/dc:DocumentCheck/dc:TimeMeasurement/dc:DocumentModelSelection/base:EndTime	The result of the model selection was processed by the IS.
DC Document Model Selection Duration	/dc:DocumentCheck/dc:TimeMeasurement/dc:DocumentModelSelection/base:Duration	DC Document Model Selection Duration := DC Document Model Selection End Time - DC Document Model Selection Start Time

Table 6.3 Timestamps and durations to be logged by an inspection system - Document Check

6.4.3 Optical check

The timestamps and durations regarding an optical check are given in ►Table 6.4.

Short Name	Relative XPath	Description
DCO Start Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:Overall/base:StartTime	The IS starts to execute optical checks.
DCO End Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the optical check.
DCO Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:Overall/base:Duration	DCO Duration := DCO End Time - DCO Start Time
DCO Processing Per Page Start Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:ProcessingPerPage/dco:Page/base:StartTime	In case the IS checks document pages optically. For each page, this process starts when the image that is used for document model identification is taken.

Short Name	Relative XPath	Description
DCO Processing Per Page End Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:ProcessingPerPage/dco:Page/base:EndTime	This point in time does not correspond to the removal of the document (or page) from the reader. Instead, it is the point in time when the optical check software has finished processing the data of this page.
DCO Processing Per Page Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:ProcessingPerPage/dco:Page/base:Duration	DCO Processing Per Page Duration := DCO Processing Per Page End Time - DCO Processing Per Page Start Time
DCO Processing Per Page Visual Zone Parsing Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:ProcessingPerPage/dco:Page/dco:VisualZoneParsingDuration	Time spent parsing the visual zone, i.e. the time between starting the OCR on the image and getting the final result. This SHALL NOT include the time spent for the OCR on the MRZ.
DCO Document Model Identification Start Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:DocumentModelIdentification/base:StartTime	The IS can try to identify the document model by visual cues. This process can start for example by analyzing the MRZ or by checking patterns. It starts when all information requested by the IS is available, i.e. after reading the MRZ and checking patterns.
DCO Document Model Identification End Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:DocumentModelIdentification/base:EndTime	The IS has determined the final result, including logging data, of the document model identification.
DCO Document Model Identification Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:DocumentModelIdentification/base:Duration	DCO Document Model Identification Duration := DCO Document Model Identification End Time - DCO Document Model Identification Start Time
DCO MRZ Correction Optical Recognition Start Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:MrzCorrectionOpticalRecognition/base:StartTime	The point in time when the IS determines that it is necessary for the user to perform a correction of the optically recognized MRZ and starts the corresponding correction process.
DCO MRZ Correction Optical Recognition End Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:MrzCorrectionOpticalRecognition/base:EndTime	The correction process has finished.
DCO MRZ Correction Optical Recognition Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:MrzCorrectionOpticalRecognition/base:Duration	DCO MRZ Correction Optical Recognition Duration := DCO MRZ Correction Optical Recognition End Time - DCO MRZ Correction Optical Recognition Start Time
DCO CAN Correction Optical Recognition Start Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:CanCorrectionOpticalRecognition/base:StartTime	The point in time when the IS determines that it is necessary for the user to perform a correction of the optically recognized CAN and starts the corresponding correction process.
DCO CAN Correction Optical Recognition End Time	/dco:OpticalCheck/dco:TimeMeasurement/dco:CanCorrectionOpticalRecognition/base:EndTime	The correction process has finished.

Short Name	Relative XPath	Description
DCO CAN Correction Optical Recognition Duration	/dco:OpticalCheck/dco:TimeMeasurement/dco:Can-CorrectionOpticalRecognition/base:Duration	DCO Document CAN Correction Optical Recognition Duration := DCO CAN Correction Optical Recognition End Time - DCO CAN Correction Optical Recognition Start Time

Table 6.4 Timestamps and durations to be logged by an inspection system - Optical Check

6.4.4 Electronic check

The timestamps and durations regarding an electronic check are given in ►Table 6.5.

Short name	Relative XPath	Description
DCE Start Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:Overall/base:StartTime	The point in time when the IS attempts to interact with the electronic component of a document. Note that this can happen at any time in the document checking process. Thus, the attempt can also fail, for example for a document without an electronic component.
DCE End Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the electronic check.
DCE Duration	/dce:ElectronicCheck/dce:TimeMeasurement/dce:Overall/base:Duration	DCE Duration := DCE End Time - DCE Start Time
DCE ProcessingDuration	/dce:ElectronicCheck/dce:TimeMeasurement/dce:ProcessingDuration	Time spent for processing the data read from the electronic component. This includes the time spent for checking the data groups and the verification of the certificates contained in the electronic component.
DCE Electronic Component Interaction Duration	/dce:ElectronicCheck/dce:TimeMeasurement/dce:ElectronicComponentInteractionDuration	Time spent interacting with the electronic component. This includes reading files and performing (the parts of) protocols that involve communicating with the electronic component.
DCE Electronic Component Feature Scan	/dce:ElectronicCheck/dce:TimeMeasurement/dce:ElectronicComponentFeatureScan	The timing information for scanning an electronic component for its features.
DCE MRZ Correction Electronic Component Access Start Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:MrzCorrectionElectronicComponentAccess/dce:Correction/base:StartTime	The point in time when the IS determines that it is necessary for the user to perform a correction of the MRZ used for the electronic component access and starts the corresponding correction process.
DCE MRZ Correction Electronic Component Access End Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:MrzCorrectionElectronicComponentAccess/dce:Correction/base:EndTime	The correction process has finished.
DCE MRZ Correction Electronic Component Access Duration	/dce:ElectronicCheck/dce:TimeMeasurement/dce:MrzCorrectionElectronicComponentAccess/dce:Correction/base:Duration	DCE MRZ Correction Electronic Component Access Duration := DCE MRZ Correction Electronic Component Access End Time - DCE MRZ Correction Electronic Component Access Start Time

Short name	Relative XPath	Description
DCE CAN Correction Electronic Component Access Start Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:CanCorrectionElectronicComponentAccess/dce:Correction/base:StartTime	The point in time when the IS determines that it is necessary for the user to perform a correction of the CAN used for the electronic component access and starts the corresponding correction process.
DCE CAN Correction Electronic Component Access End Time	/dce:ElectronicCheck/dce:TimeMeasurement/dce:CanCorrectionElectronicComponentAccess/dce:Correction/base:EndTime	The correction process has finished.
DCE CAN Correction Electronic Component Access Duration	/dce:ElectronicCheck/dco:TimeMeasurement/dce:TimeMeasurement/dce:CanCorrectionElectronicComponentAccess/dce:Correction//base:Duration	DCE Document CAN Correction Electronic Component Access Duration := DCE CAN Correction Electronic Component Access End Time - DCE CAN Correction Electronic Component Access Start Time
DCE Task Start Time	/dce:ElectronicCheck/dce:ProcessInformation/dce:Trace/dce:Task/@startTime	The start time of a task.
DCE Task End Time	/dce:ElectronicCheck/dce:ProcessInformation/dce:Trace/dce:Task/@endTime	The end time of task.
DCE Apdu Start Time	/dce:ElectronicCheck/dce:ProcessInformation/dce:Trace/dce:Task/dce:Apdus/dce:Apdu/@startTime	The start time of an Application Protocol Data Unit (APDU) transmission.
DCE Apdu End Time	/dce:ElectronicCheck/dce:ProcessInformation/dce:Trace/dce:Task/dce:Apdus/dce:Apdu/@endTime	The end time of an APDU transmission.

Table 6.5 Timestamps and durations to be logged by an inspection system - Electronic Check

6.4.5 Seal check

The timestamps and durations regarding a seal check are given in ►Table 6.6.

Short name	Relative XPath	Description
DCS Start Time	/dcs:SealCheck/dcs:TimeMeasurement/dcs:Overall/base:StartTime	The point in time when the IS starts an optical scan in order to detect a digital seal.
DCS End Time	/dcs:SealCheck/dcs:TimeMeasurement/dcs:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the seal check.
DCS Duration	/dcs:SealCheck/dcs:TimeMeasurement/dcs:Overall/base:Duration	DCS Duration := DCS End Time - DCS Start Time
DCS Detection Duration	/dcs:SealCheck/dcs:TimeMeasurement/dcs:DetectionDuration	Time spent determining whether a digital seal is present on the document.
DCS Scan Duration	/dcs:SealCheck/dcs:TimeMeasurement/dcs:ScanDuration	Time spent translating the physical representation of the digital seal into a digital representation.
DCS Check Duration	/dcs:SealCheck/dcs:TimeMeasurement/dcs:CheckDuration	Time spent checking the digital representation of the seal.

Table 6.6 Timestamps and durations to be logged by an inspection system - Seal Check

6.4.6 Combined check

The timestamps and durations regarding a combined check are given in ►Table 6.7.

Short name	Relative XPath	Description
DCC Start Time	/dcc:CombinedCheck/dcc:TimeMeasurement/dcc:Overall/base:StartTime	A combined check can be performed when information from partial checks becomes available. Hence, it is not necessary that all partial checks are finished for the combined checks to start. However DCC Start Time must be later than the start time of at least one partial check. DCC Start Time > one of [DCE Start Time, DCO Start Time, DCS Start Time]
DCC End Time	/dcc:CombinedCheck/dcc:TimeMeasurement/dcc:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the combined check.
DCC Duration	/dcc:CombinedCheck/dcc:TimeMeasurement/dcc:Overall/base:Duration	DCC Duration := DCC End Time - DCC Start Time

Table 6.7 Timestamps and durations to be logged by an inspection system - Combined Check

6.4.7 Cross document check

The timestamps and durations regarding a cross document check are given in ▶Table 6.8.

Short name	Relative XPath	Description
XDC Start Time	/xdc:CrossDocumentCheck/xdc:TimeMeasurement/xdc:Overall/base:StartTime	The point in time when the IS starts to compare the two documents that are subject to the cross document check.
XDC End Time	/xdc:CrossDocumentCheck/xdc:TimeMeasurement/xdc:Overall/base:EndTime	The point in time when the IS produces the final result, including logging data, of the cross document check.
XDC Duration	/xdc:CrossDocumentCheck/xdc:TimeMeasurement/xdc:Overall/base:Duration	XDC Duration := XDC End Time - XDC Start Time

Table 6.8 Timestamps and durations to be logged by an inspection system - Cross Document Check

6.4.8 Performance capabilities

It is RECOMMENDED to minimize the total duration of the document inspection without reducing the quality of the result. Each inspection step SHOULD be executed as soon as the input data, which is mandatory for the step is available to the IS, for example, a digitized MRZ or CAN to access the electronic component of the document under inspection. By passing the MRZ/CAN data to the electronic check component of the IS at an early stage, the total time of the inspection could be reduced, since the electronic and optical check component can perform their checks simultaneously. The following diagram (▶Figure 6.1) illustrates the temporal relation between the optical, electronic, digital seal and combined inspection of the document.

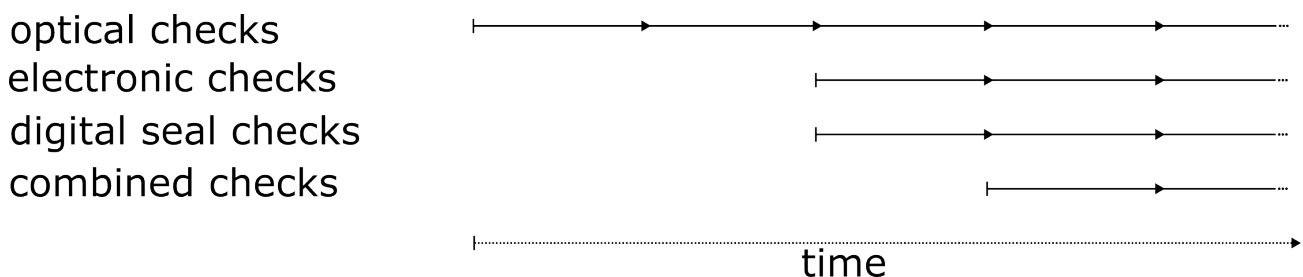


Figure 6.1. Temporal relation of optical, electronic, digital seal and combined checks

7 Conformity

In order to conform to this Technical Guideline, an Inspection System and Inspection Application SHALL completely implement and meet all normative requirements set out in it, as well as the profiling defined by the public agency for the scope of application.

In addition, the document readers used SHALL be certified according to [BSI TR-03105-4], [BSI TR-03105-5.1] and [BSI TR-03105-5.2], and meet the requirements specified therein.

List of Abbreviations

Abbreviation	Description
AA	Active Authentication
APDU	Application Protocol Data Unit
BAC	Basic Access Control
BKA	Bundeskriminalamt
BPMN	Business Process Model and Notation
BPOL	Bundespolizei
BSI	Bundesamt für Sicherheit in der Informationstechnik
BVA	Bundesverwaltungsamt
CA	Chip Authentication
CAN	Card Access Number
CSCA	Country Signing Certificate Authority
CVCA	Country Verifying Certificate Authority
DF	Dedicated File
DN	Distinguished Name
DS	Document Signer
DTC	Digital Travel Credential
DTC-PC	DTC Physical Component
DTC-VC	DTC Virtual Component
DV	Document Verifier
EAC	Extended Access Control
eID	Electronic Identity Document
eMRTD	Electronic Machine Readable Travel Document
FAIL	Forensic Analysis Information List
ICAO	International Civil Aviation Organization
IR	infrared
IS	Inspection System
ISO	International Organization for Standardization
JPEG	JPEG
MRTD	Machine Readable Travel Document
MRZ	Machine Readable Zone
OCR	Optical Character Recognition
PA	Passive Authentication
PACE	Password Authenticated Connection Establishment
PACE-CAM	Password Authenticated Connection Establishment - Chip Authentication Mapping
PKD	Public Key Directory
PKI	Public Key Infrastructure
PNG	Portable Network Graphics
PPI	pixels per inch
RF	Radio-Frequency

Abbreviation	Description
RFID	Radio-Frequency Identification
SPOC	Single Point of Contact
TA	Terminal Authentication
TR	Technical Guideline
USB	Universal Serial Bus
UUID	Universally Unique Identifier
UV	ultraviolet
VI	white light
VIZ	Visual Zone
XML	Extensible Markup Language
XSD	XML Schema Definition

Bibliography

- [BSI TR-03105-4] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie TR-03105 Part 4 – Test Plan for ICAO-compliant Proximity Coupling Devices (PCD) on Layers 1-4, 2016, Version 3.0.
- [BSI TR-03105-5.1] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie BSI TR-03105 Part 5.1 – Test plan for ICAO compliant Inspection Systems with EACv1, 2016, Version 1.4.1.
- [BSI TR-03105-5.2] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie BSI TR-03105 Part 5.2 – Test plan for eID and eSign compliant smart card readers with integrated EACv2, 2013, Version 1.1.
- [BSI TR-03110] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie TR-03110 – Advanced Security Mechanisms for Machine Readable Travel Documents, Part 1 & 3, 2012, Version 2.10.
- [BSI TR-03121] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie TR-03121 - Biometrics in public sector applications.
- [BSI TR-03127] BSI TR-03127 Architecture electronic Identity Card and electronic Resident Permit (informative translation version, 2011, Version 1.13.
- [BSI TR-03129-2] Protocols for the Management of Certificates and CRLs in Public-Key-Infrastructures (PKIs) - Part 2: Supplemental specifications for public and official authorities, Version 1.4.1.
- [BSI TR-03135] BSI TR-03135, Technical Guideline Machine Authentication of MRTDs for Public Sector Applications, Version 5.0.1.
- [BSI TR-03135-2] BSI TR-03135-2, Technical Guideline Machine Authentication of MRTDs for Public Sector Applications. Part 2: Application profiles for official document inspection systems, Version 5.0.1.
- [BSI TR-03156] Bundesamt für Sicherheit in der Informationstechnik (BSI): Technische Richtlinie TR-03156 - Hoheitliches Identitätsmanagement mit EU-Informationssystemen.
- [FRONTEX] European Agency for the Management of Operational Cooperation at the External Borders of the Member States of the European Union (FRONTEX): “Best Practice Technical Guidelines for Automated Border Control (ABC) Systems”, 2012; “Best Practice Operational Guidelines for Automated Border Control (ABC) Systems”, 2015.
- [ICAO9303] ICAO Doc 9303 – Machine Readable Travel Documents, 8th Edition, 2021.
- [ICAODTC-PC] International Civil Aviation Organization (ICAO): Technical Report – Digital Travel Credentials (DTC) - Physical Component and Protocols, 2022.
- [ICAODTC-VC] International Civil Aviation Organization (ICAO): Technical Report – Digital Travel Credentials (DTC) - Virtual Component Data Structure and PKI mechanisms, 2024.
- [ISO/IEC 30116:2016] Information technology – Automatic identification and data capture techniques – Optical Character Recognition (OCR) quality testing, Edition 1, 2016.
- [ISO/IEC-39794-5] ISO/IEC 39794-5: Information technology – Extensible biometric data interchange formats – Part 5: Face image data, 2019.
- [ISO10918-1] ISO/IEC 10918-1: Digital compression and coding of continuous-tone still images: Requirements and guidelines, Edition 1, 1994.
- [ISO14443-2] ISO/IEC 14443: Identification cards -- Contactless integrated circuit cards -- Proximity cards -- Part 2: Radio frequency power and signal interface, Edition 3, 2016.
- [ISO1831] ISO 1831: Printing specifications for optical character recognition, Edition 1, 1980.
- [ISO3166-1] ISO 3166: Codes for the representation of names of countries and their subdivisions – Part 1: Country codes, Edition 3, 2013,
- [ISO3166-2] ISO 3166: Codes for the representation of names of countries and their subdivisions – Part 2: Country subdivision code, Edition 3, 2013.
- [KeepAChangelog] Keep a Changelog – <https://keepachangelog.com/en/1.0.0/>.

- [PRADO] *The General Secretariat of the Council: PRADO - Public Register of Authentic travel and identity Documents Online*, <http://www.consilium.europa.eu/prado/en/prado-start-page.html/>.
- [RFC2119] *Request For Comments Editor (RFC)*, Bradner, Scott: *Key words for use in RFCs to indicate requirement levels*, RFC 2119, 1997.
- [RFC4122] *Request For Comments Editor (RFC)*, Leach P., Mealling M., Salz R.: *A Universally Unique Identifier (UUID) URN Namespace*, RFC 4122, 2005.